

# A Benchmark and Diagnostic Study of Epistemic Admission in Shared Agent Memory

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## Abstract

Evaluating claim admission in shared agent memory is challenging because repeated claims may be mistaken for independent evidence. An agent may copy or paraphrase a retrieved belief, while admitting a false claim exposes subsequent agents to it. To study this problem, we introduce the Correlated Promotion Benchmark (CPB), which evaluates whether candidate claims should be admitted to shared memory. CPB-Static constructs a frozen test split from publicly annotated sources with fixed gold actions. CPB-Live runs multi-agent teams over a shared store, records all writes and retrievals, and tracks source lineage defined by each scenario. A separate consumer answers from the store alone. We evaluate eight admission policies across four agent families. Our results show that policies which deduplicate sources reject many true claims alongside false ones, whereas policies preserving answer coverage admit nearly as many false claims as unrestricted sharing. Gating on declared source type reduces false adoption to 0.06–0.09, compared with 0.22–0.47 for other answering policies. Once an uncontested false belief enters memory, the consumer asserts it in 0.97–0.99 of probes across all families. No non-oracle policy consistently rejects false claims across verbatim copies, paraphrases, and paraphrases declared authoritative. These findings reveal the limitations of admission policies without access to source lineage.

## 1 Introduction

Shared agent memory poses two problems single-agent memory does not. An agent that restates a retrieved belief adds agreement without independent evidence, which can reinforce false beliefs and yield spurious majorities (Xu et al., 2026; Lin et al., 2026; Qi et al., 2026), and the cost of an error grows with the team that later reads it (Rahman et al., 2026; Papadopoulos et al., 2026). Existing benchmarks rarely hold conflicting or unequally reliable evidence (Singh, 2026), and no fixed corpus can measure the feedback between admission, retrieval and later writes. That takes a running system with logged retrieval and source lineage fixed independently of the policy.

The problem is not hypothetical. Agent frameworks ship shared memory tiers with provenance and access control (Rezazadeh et al., 2025; Margalit et al., 2026; Sidik and Rokach, 2026; Ren et al., 2026), a deployed layer deduplicates on every write (Chhikara et al., 2025), admission is defended by consistency checks, rules over correlated traces or declared derivation (Wei et al., 2025; Qi et al., 2026; Li et al., 2026b), one injected memory steers the agents that read it (Chen et al., 2024; Dong et al., 2025), and agreement among copying sources carries no evidence (Dong et al., 2009b; 2010). None of these measures what an admitted falsehood does once agents read, restate and write it back.

We introduce the Correlated Promotion Benchmark, CPB. CPB-Static combines verdicts from public annotated sources with an authored multi-agent envelope and scores policies at scale against a fixed gold action. CPB-Live runs a team over a shared store that logs every write and retrieval, with lineage fixed by the scenario, exposure read from the log and adoption graded from consumer answers, so the consequence of admitting a claim is observed rather than assumed (Dong et al., 2009b; Qi et al., 2026; Lin et al., 2026).

Across voting, lineage-collapse, confidence, source-type and judge policies on four agent families, three patterns recur. Consumers relying only on shared memory almost always repeat an uncontested false belief. Policies that refuse copies refuse most true claims with them, policies that answer admit most of what share-all admits, the judges that filter halve damage at best, and the one rule between them gates on a source-type field.

Three contributions follow.

**An instrument:** CPB, an assembly protocol over public annotated sources with its adapters, gold-action mapping and frozen test split, and CPB-Live, a running team over a store that logs every write and retrieval, so that exposure and adoption are observed after each admission and lineage is fixed by the scenario rather than inferred by the policy.

**A diagnosis of admission mechanisms:** on four agent families, reimplemented voting, collapse and judge mechanisms and components from Mem0 and A-MemGuard admit false claims under rewording or authoritative source typing, and only the governance rule, a source-type gate, keeps false adoption low while still answering.

**Measurements of the store itself:** how often a consumer adopts a false shared belief, what a competing truth changes, and how rewording and arrival order move admission, whichever mechanism is preferred.

## 2 Related Work

**Copy-aware truth discovery.** DEPEND discounts the vote of a source identified as a copier, and later work extends the model to update histories, complex copying and correlated fusion (Dong et al., 2009a;b; 2010; Pochampally et al., 2014), all inferring copying from statistical agreement since derivation is not observed.

**Admission to agent memory.** Single-writer systems gate entry by factual support, self-consistency, utility or transition faithfulness, or choose among persisting, verifying and asking (Zhang and Li, 2026; Zhang et al., 2026; Li et al., 2026a; Xiao et al., 2026; Yang et al., 2026; Hu et al., 2026; Latimer et al., 2025), without modelling a shared tier and its readers. Learning to Share admits entries to a memory shared across teams by learned utility (Fioresi et al., 2026), other systems supply shared tiers, provenance, access control and governance primitives (Rezazadeh et al., 2025; Margalit et al., 2026; Ren et al., 2026; Sidik and Rokach, 2026; Han et al., 2026; Zhang et al., 2025; Wu et al., 2026; Bhardwaj et al., 2026; Ding et al., 2026; Wang et al., 2026), and selection for shared institutional state has been framed as governance (Cuadros et al., 2026). GovMem discounts correlated traces under synthetic or inferred dependencies and specifies its harm experiment without executing it (Qi et al., 2026), MemTX commits beliefs over a writer-declared derivation graph with rollback (Li et al., 2026b), CAMA decouples correlated memories at retrieval and names the write-back loop without mitigating it (Lin et al., 2026), and poisoning defences target adversarial injection (Wei et al., 2025; Chen et al., 2024; Dong et al., 2025). Mem0’s write path and A-MemGuard’s consistency check run inside CPB-Live, Section 5.2.

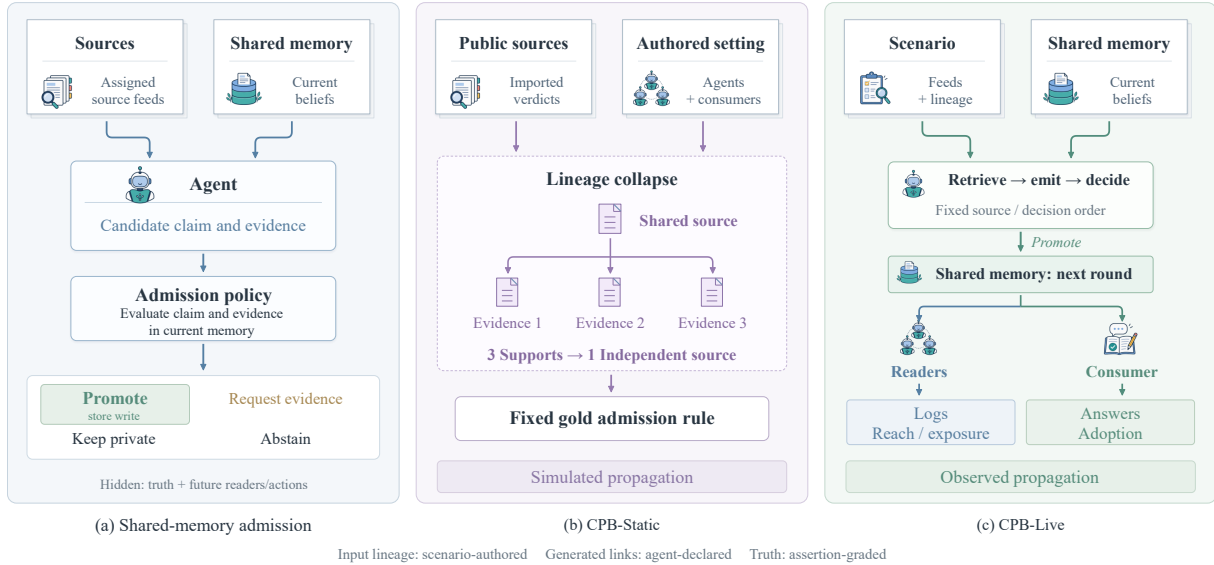


Figure 1: The instrument. Left: a policy chooses one of four actions for a candidate and its evidence, and only promote writes the store. Middle: CPB-Static collapses lineage, so three supports from one source count as one, and scores against a fixed gold rule with simulated propagation. Right: CPB-Live reads reach and exposure from the store’s log and adoption from the consumer’s answer.

**Propagation and benchmarks.** Multi-agent systems adopt false beliefs collectively and consolidate them into memory (Xu et al., 2026), harm amplifies under perturbation and ideas spread through interaction alone (Rahman et al., 2026; Papadopoulos et al., 2026), majorities can correct misinformed agents (Becker et al., 2026), judge panels err together (Kohli, 2026), and belief-based memory helps most under conflict (Singh, 2026). Risk-weighted gating, abstention and stale inherited constraints are studied without shared memory (Iyer and Babu, 2026; Liu et al., 2026; Nakayashiki, 2026). Memory benchmarks test retention and retrieval by one writer (Maharana et al., 2024; Wu et al., 2025; Hu et al., 2025), conflict benchmarks how one model resolves contradictions (Su et al., 2024; Xie et al., 2024; Xiang et al., 2026), and belief revision, partial observability and claim-level correction have their own lines (Park, 2026; Liao et al., 2026; Kong et al., 2026). None executes admission in a running team with logged shared-memory access and measured adoption.

## 3 Method

### 3.1 Problem formulation

CPB measures admission to shared memory and its consequences, Figure 1. A team  $\mathcal{A}$  shares beliefs  $S_t$  at round  $t$ . Each agent observes its assigned sources, retrieves  $S_t$  and emits candidate claims, each candidate  $m$  with evidence  $E_m$ . An admission policy  $\pi$  maps  $\langle m, E_m, S_t \rangle$  to PROMOTE, REQUEST\_EVIDENCE, KEEP\_PRIVATE or ABSTAIN, and only PROMOTE changes the store, making the claim retrievable by every agent from the next round.

Candidates are processed in a fixed agent order within each round. Every policy receives the same scheduled sources, but later candidates differ because agents retrieve different admitted beliefs. The policy sees neither claim truth nor its future readers.

### 3.2 Measurement requirements

**Distinguishing independent support from repetition.** Evaluating dependence needs a copy relation independent of the policy under test.  $E_m$  is an evidence graph whose nodes carry a stance, a source identifier and, where available, a parent source. Lineage collapse merges nodes that share a parent or derive from one another and counts the remaining supporting groups, as copy-aware truth discovery does with inferred parents (Dong et al., 2009b). Inferred parents need many sources and confound copying with correlated truth, and the dependencies GovMem and MemTX act on are synthetic or writer-declared (Qi et al., 2026; Li et al., 2026b). CPB fixes the relation in the scenario, so a policy’s collapse is scored against it and not against a similarity proxy.

In CPB-Live the scenario fixes the lineage of every feed before any policy runs, a reference that owes nothing to text similarity. Generated claims are another matter: the store logs retrieval but not which retrieved belief a candidate restates, so that derivation is available only through agent declarations, whose incompleteness we measure.

**Observing downstream exposure.** The consequences of a wrong promotion depend on who retrieves it and what they do with it. Assigning reach and severity in advance specifies a cost model without measuring a consequence, so CPB-Live records retrievals after admission and probes consumer answers: the log gives reach and exposure within the simulation, the probe adoption. GovMem specifies this experiment and CAMA names the loop it closes (Qi et al., 2026; Lin et al., 2026), and CPB-Live executes it.

### 3.3 The instrument

**CPB-Static** is an assembly protocol over public annotated sources. It imports verdicts and adds an authored multi-agent setting: agent observations, expected reach, severity and consumer tasks. A fixed rule maps each episode to a gold action, promoting a supported claim with enough independent sources and withholding otherwise. Since the rule counts sources after lineage collapse and simulates propagation over authored consumers, Static scores agreement with this specification and not the emergent effects of dependence.

**CPB-Live** runs the team of Section 3.1 over a store that logs every write and retrieval, and a consumer then answers a targeted question from what the store holds. Reach and exposure are read from the log, adoption from the consumer’s answer and lineage from the scenario, so both conditions of Section 3.2 are met.

Let  $F$  be the admitted beliefs the grader marks false,  $t_b$  the round at which belief  $b$  was written,  $R(b)$  the agents that retrieved it, and  $C$  the consumer’s reads, each a round  $r$ . Realised reach is  $|\bigcup_{b \in F} R(b)|$ . Propagation damage is  $|\{r \in C : \exists b \in F, t_b < r\}|$ , the consumer reads made while a false belief was retrievable, which is exposure within the simulation and not real-world harm. Correlated agreement is the number of candidates whose declaration names a shared belief, amplification the number whose declaration names a belief in  $F$  divided by  $|F|$ , and order sensitivity the share of swapped-order pairs whose false or true write counts differ.

Claim truth is judged apart from provenance: a fixed grader decides whether a candidate’s sentence asserts the scenario’s false or true proposition, so an uncited restatement counts as false and a hedge citing a false source does not, while citations and declared dependencies describe provenance alone. Appendix N compares the rules and Appendix L validates the grader.

## 4 Experimental Setup

### 4.1 CPB-Static

CPB-Static imports verdicts from claim-verification corpora (Schlichtkrull et al., 2023; Thorne et al., 2018), supersession chains from a multi-hop editing corpus (Zhong et al., 2023), and lineage groups from the Wikipedia citation graph (Petroni et al., 2022). Two annotators assess under a released protocol whether each group asserts one proposition. The 1,200 six-agent episodes comprise 600 training, 200 validation and 400 test episodes. The gold rule requires two independent sources, and the test split was hashed and frozen before evaluation. Appendix B details construction, audit and provenance.

### 4.2 CPB-Live

Each episode runs six agents for six rounds over one configuration committed in advance, so episodes and configurations count the same unit, and all scenario content is authored fiction.

Four scenario families test distinct interactions. **Correlated agreement** gives one agent a root fact and, in half the scenarios, a second agent independent corroboration. **Concurrent conflict** gives two agents incompatible evidence simultaneously, followed by authoritative confirmation two rounds later. **Mixed** scenarios pair a background fact with a later feed requiring that context. **Dual source** gives one agent a true root with an independent restatement or a false root with a verbatim or reworded copy in the same round.

Each backbone runs 100 configurations: 60 base configurations, 50 scenarios and their 10 order-swapped variants, 20 dual-source configurations split between verbatim copies and independent restatements, and 20 reworded-copy configurations split between web and authoritative source types. Thirty more complete the cross of truth and form, ten each with a true root beside a verbatim copy, a true root beside a reworded copy, and a false root beside an independent restatement typed authoritative. They run on Qwen and the closed family and never enter the pooled rows of the eight policies. The judge comparison uses the 60 base configurations, and the consumer probe of Section 3.3 is asked at round six over the committed transcript.

### 4.3 Policies and backbones

Ten policies run on Static and eight on Live. Private only and share-all are the endpoints. Confidence gating thresholds a proxy, and majority and unanimity votes count nominal support, over agent stances on Static and over cited sources on Live, where the vote promotes on one supporting citation with no contradiction, keeps private when contradictions outnumber supports, and requests evidence otherwise. The independence vote requires two supporting groups after near-duplicate collapse. DEPEN discounts suspected copying from statistical agreement and source accuracy (Dong et al., 2009b), and GovMem’s review-all-correlated rule routes correlated support to review (Qi et al., 2026). Governance, the source-type gate, promotes fresh authoritative support without contradiction, abstains on an unresolved contradiction and otherwise defers. The LLM judge returns one action from a textual view of the same input, and logistic regression and gradient-boosted trees are trained baselines over shared structured features.

Policies receive the same reconstructed public view without the retrieval log, and equal validation budgets for tuning. Live rules use candidate-level cited evidence rather than fixed agent votes, Appendix C, and confidence gating, unanimity and the trained baselines run on Static only. Two diagnostic arms of the independence vote run on the dual-source sets of Qwen and the closed family, with their own rows pooled over the two dual-source sets every family ran, the cross set entering the cross table alone: a lineage oracle whose copy groups are the authored lineage, information no policy can have, and a semantic collapse whose

copy groups are components of bidirectional entailment under the pinned scorer. A share-all arm with the agents’ retrieval of shared memory switched off runs on the base configurations of the same two families. Live also runs components from two published systems, Section 5.2: Mem0 as the store, its own write path deciding what an admitted candidate becomes (Chhikara et al., 2025), and A-MemGuard’s consistency check as a policy over the candidate, its cited sources and the nearest visible beliefs (Wei et al., 2025). Table 2 in Appendix A lists coverage.

Qwen3.5-9B is the primary backbone and Gemma-3-12B and Llama-3.1-8B replicate it (Qwen Team, 2026; Kamath et al., 2025; Grattafiori et al., 2024), all three commit-pinned and decoding greedily with reasoning disabled, and Claude Opus 5, the closed family, uses provider-default sampling through the Anthropic API with every response persisted (Anthropic, 2026). Each family’s backbone serves as its six agents, consumer and judge, Claude also judges Qwen on the 60 base configurations, and the selected Static judge prompt runs on all four backbones.

## 4.4 Metrics and statistics

Static false-promotion rate is the proportion promoted among candidates with a non-promote gold action. Live adoption is the proportion of probes asserting the false value, and correctness is the proportion asserting the true value among scenarios containing one. Candidate truth, exposure and provenance follow Section 3.3, and Appendices L and M validate the two graders.

Static intervals use source-item cluster bootstraps, six comparisons specified before test access use Holm correction, and every other analysis, Live included, is pre-registered as descriptive and reported without hypothesis tests.

## 5 Results

Ten policies run on the frozen Static split and eight on Live, plus a frontier judge over Qwen agents, on four agent families with one hundred configurations each, Table 2, under the metrics of Sections 3.3 and 4.4. Correction, arrival order and declared provenance are in Appendix K and the pre-registered Static comparisons in Appendix Q.

### 5.1 Admission, exposure and downstream utility

**F1. Among the tested policies, only governance keeps damage low while still answering in every family, and the judges that filter halve damage at best.**

Table 1 reports the three rates over each policy’s own stream, where private only is 0.000 and share-all 1.000 by construction and 1.000 elsewhere means every candidate of that kind was admitted, and Figure 2 adds the probe’s adoption and correctness, Appendix E.

Majority vote matches share-all on Qwen in damage and FA, since candidates cite supporting evidence without exposing contradictions, and the few true candidates it withheld arrived with a contradiction in view. On Gemma and Llama its damage stays at 0.985 and 0.981 while FA falls to 0.649 and 0.108, since one admitted copy exposes the same readers, and only on the closed family does damage fall, to 0.864.

The local judge matches share-all on Qwen and nearly so on Gemma, at FA 0.988. The Llama and closed-family judges halve damage, to 0.483 and 0.472, at FA 0.325 and 0.473 and TA 0.323 and 0.435. Rejection

Table 1: CPB-Live, the 100 configurations of each family pooled. **Dmg**: damage as a share of share-all’s. **FA** and **TA**: false and true admissions as shares of the false and true candidates the policy saw. The last block is the two published systems of Section 5.2.

| Policy          | Qwen3.5-9B |       |       | Gemma-3-12B |       |       | Llama-3.1-8B |       |       | Claude Opus 5 |       |       |
|-----------------|------------|-------|-------|-------------|-------|-------|--------------|-------|-------|---------------|-------|-------|
|                 | Dmg↓       | FA↓   | TA↑   | Dmg↓        | FA↓   | TA↑   | Dmg↓         | FA↓   | TA↑   | Dmg↓          | FA↓   | TA↑   |
| Private only    | 0.000      | 0.000 | 0.000 | 0.000       | 0.000 | 0.000 | 0.000        | 0.000 | 0.000 | 0.000         | 0.000 | 0.000 |
| Share all       | 1.000      | 1.000 | 1.000 | 1.000       | 1.000 | 1.000 | 1.000        | 1.000 | 1.000 | 1.000         | 1.000 | 1.000 |
| Majority vote   | 1.000      | 1.000 | 0.964 | 0.985       | 0.649 | 0.459 | 0.981        | 0.108 | 0.100 | 0.864         | 0.672 | 0.529 |
| LLM judge       | 1.000      | 1.000 | 1.000 | 0.985       | 0.988 | 0.975 | 0.483        | 0.325 | 0.323 | 0.472         | 0.473 | 0.435 |
| Governance rule | 0.152      | 0.161 | 0.481 | 0.137       | 0.145 | 0.463 | 0.151        | 0.137 | 0.133 | 0.112         | 0.120 | 0.348 |
| Independence    | 0.289      | 0.306 | 0.127 | 0.244       | 0.258 | 0.123 | 0.257        | 0.119 | 0.051 | 0.243         | 0.250 | 0.108 |
| Mem0 store      | 1.000      | 1.000 | 1.000 | 1.000       | 1.000 | 1.000 | 0.940        | 1.000 | 1.000 | 0.991         | 1.000 | 1.000 |
| Mem0 gated      | 0.152      | 0.161 | 0.481 | 0.137       | 0.083 | 0.373 | 0.151        | 0.079 | 0.112 | 0.112         | 0.120 | 0.337 |
| A-MemGuard      | 1.000      | 1.000 | 0.982 | 0.985       | 0.988 | 0.869 | 0.966        | 0.819 | 0.791 | 0.995         | 1.000 | 0.667 |

alone does not settle utility: the Llama judge reaches 0.78 correctness, the highest on its family, at 0.22 adoption, and the closed-family judge 0.40 at 0.19. On Static the frontier judge has the lowest false-promotion rate, 0.077 against 0.228 to 0.285, and the lowest accuracy, 0.41, since it keeps 0.54 of candidates private against 0.00 to 0.31 for the open judges: at its one operating point it withholds more rather than separating better, and equal coverage is not tested here, Appendix O.

The independence class, on its merged row, admits 0.119 to 0.306 of false candidates and 0.051 to 0.127 of true ones, so it rejects true claims more firmly than false ones on every family, with damage shares of 0.243 to 0.289, correctness of 0.13 to 0.17 and adoption of 0.11 to 0.18. Its three rules agree on the open families under one collapse predicate and differ on the closed family, Appendix C.

The governance rule holds damage shares of 0.112 to 0.152, FA of 0.120 to 0.161 and TA of 0.133 to 0.481, with 0.06 to 0.09 adoption and 0.58 to 0.65 correctness. Its correctness is 0.90 on concurrent conflict, 0.90 on correlated agreement and 0.00 on mixed scenarios, so it is strongest where the scenario supplies the authoritative source type it gates on, which Section 5.5 tests directly.

## 5.2 Two published components inside the instrument

**F2. Mem0 removes duplicates and not false adoption, and A-MemGuard’s check passes falsehoods their cited sources support.**

The last block of Table 1 gives the two components’ admission rates, and Appendix D what the store held and what the consumer answered. Under share-all Mem0 rejects nothing: its damage share is 1.000 on Qwen and Gemma, 0.991 on the closed family and 0.940 on Llama, where absorbed re-raising delay some first false writes by a round, it holds a false belief in 0.62, 0.62, 0.60 and 0.50 of configurations on Qwen, Gemma, Llama and the closed family against the native store’s 0.62, 0.62, 0.63 and 0.50, and adoption is 0.51, 0.32, 0.41 and 0.43 against 0.45, 0.32, 0.47 and 0.44. Deduplication absorbs 0.67 of Gemma’s admissions and 0.86 of Llama’s, mostly by folding false candidates into false beliefs already held, so it removes redundancy and not error.

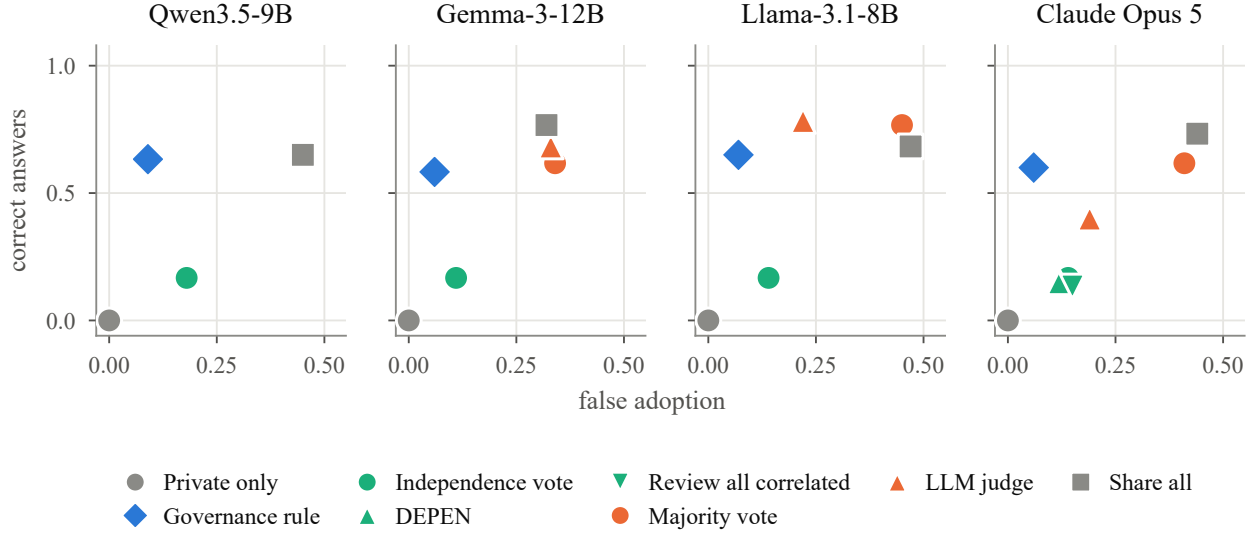


Figure 2: False adoption and correctness on CPB-Live, the eight policies on the 100 configurations of each family. Values in Appendix E, where the diagnostic arms and the frontier judge over Qwen appear on their own pools.

Behind the governance rule Mem0 holds the rule’s own 0.10, 0.09, 0.10 and 0.06, is adopted at 0.09, 0.06, 0.07 and 0.07, and still admits the authoritatively typed rewording.

A-MemGuard’s consistency check rejects 0.00, 0.01, 0.18 and 0.00 of false candidates, with damage shares of 1.000, 0.985, 0.966 and 0.995, false beliefs held in 0.62, 0.61, 0.61 and 0.50 of configurations and adoption of 0.45, 0.33, 0.48 and 0.45. On the closed family it keeps about half of all candidates private yet rejects no false one, since each arrives with supporting source text, so what it withholds is a third of the true candidates. The tested write and consistency checks leave falsehoods supported by their cited evidence admissible.

### 5.3 Adoption with and without a competing truth

**F3. With shared memory as the only source, an uncontested false belief is asserted in 0.97 to 0.99 of probes, and adoption is lower beside a competing truth.**

The consumer has no source of its own, and it abstained in all 1,551 probes whose episode promoted nothing. Figure 3 splits the probes with exactly one visible false belief by whether a true belief stood beside it. Without one, false adoption was 0.98, 0.99, 0.99 and 0.97 on Qwen, Gemma, Llama and the closed family, 553 probes in all. Under this constraint, every model repeats an uncontested false memory as fact.

With a competitor, answers naming the false value alone fall to 0.15 of 160 probes, and 46 more name both values. Counting those, the false value is voiced in 0.60 of Qwen probes, 0.43 of Gemma, 0.33 of Llama and 0.16 of closed-family probes, so a competing truth is where the models part.

All 192 probes with both a true and a false belief visible come from concurrent-conflict scenarios, the only family that delivers a later confirmation, so the protection is measured there alone, and it fades with accumulation: with three or more false beliefs beside one true one, false-only adoption returns to 0.42, Appendix F.

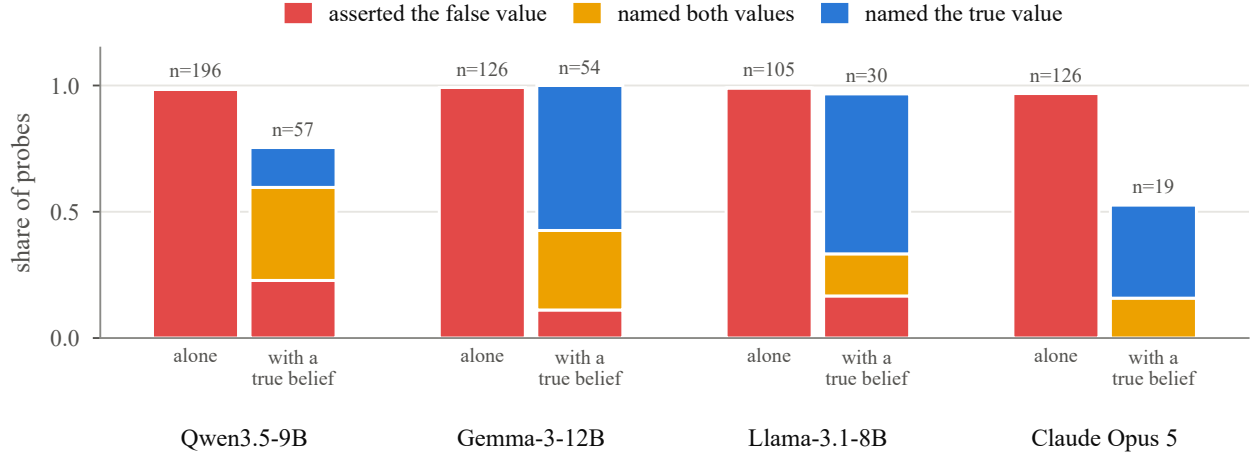


Figure 3: Consumer answers with exactly one visible false belief, the nine policies’ probes on the pool of Table 5, by family. Left bars show stores without a competing true belief, and right bars include one. Answers naming both values count as voicing the false value.

#### 5.4 Refusal on each policy’s own candidate stream

**F4. Governance refuses most false candidates in every family, while majority-vote refusal runs from zero to nine in ten with the backbone.**

Under share-all the agents emit 0.62, 0.81, 7.35 and 0.56 false candidates per configuration over the 100 configurations on Qwen, Gemma, Llama and the closed family, so Table 8 in Appendix G conditions on a false candidate and reports the share refused, drawn as Figure 5. The governance rule refuses 0.84 to 0.88 and the independence class 0.67 to 0.88.

Majority vote refuses 0.00 on Qwen, 0.35 on Gemma, 0.89 on Llama and 0.33 on the closed family. The live vote promotes when a cited source supports the claim and nothing visible contradicts it, so its refusal follows what the agents cite, and most of Llama’s false candidates are uncited restatements. Local judges range from 0.00 to 0.68.

Across families the rule’s refusal moves by 0.04 and the vote’s by 0.89, since the rule gates on a source type every stream carries and the vote on evidence whose amount varies with the family.

#### 5.5 Sensitivity to rewording and source type

**F5. Among the tested promoting policies, only the lineage oracle refuses all three copy regimes. Rewording passes surface collapse and authoritative typing passes the governance rule.**

Dual-source configurations give one agent two sources in one round, Figure 4 and Appendix H. The independence class refuses the verbatim copy in every family, at 0.00 adoption, and admits the reworded copy, which states the same proposition below the similarity threshold, in most configurations, at 0.65 to 0.72 adoption pooled over families.

The governance rule holds 0.00 adoption on both copies while they are typed as web text and 0.70 once the reworded copy is typed as a register document, so its protection rests on source type and not on detecting the copy. Summed over policies, false admissions rise from 12 to 30 on verbatim copies to 38 to 54 on reworded and 43 to 70 on authoritatively typed ones.

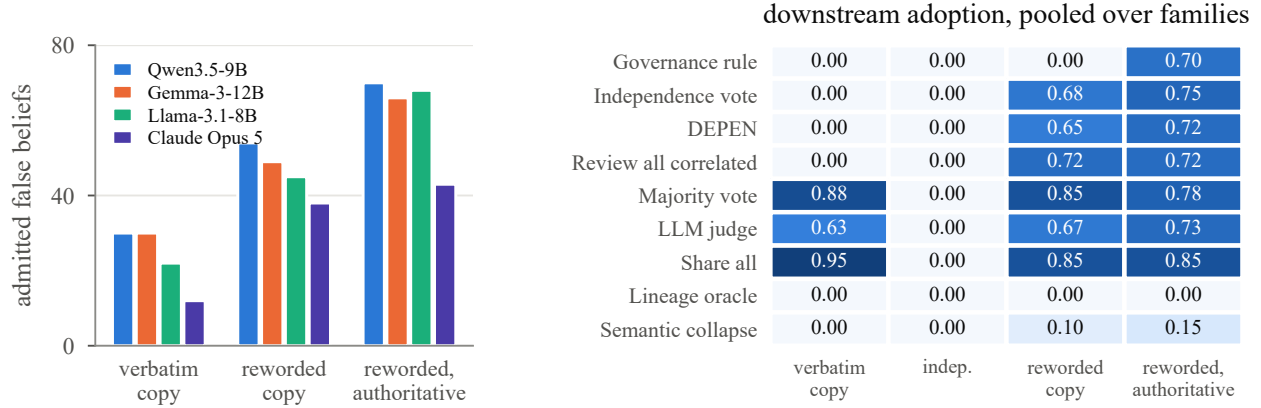


Figure 4: Dual-source copy regimes. Left: false admissions summed over the nine policies, by regime and family. Right: downstream adoption by policy and regime, pooled over the families’ configurations.

## 5.6 Independence against truth

**F6. Admission follows the form of the second source more than the truth of the claim: no policy refuses a verbatim copy of one truth and admits it of the other, and whatever admits a false independent restatement admits the true one.**

The cross set adds the missing cells, a true root beside a verbatim copy, a true root beside a reworted copy and a false root beside an independent restatement typed as a register document, ten each on Qwen and the closed family, and Table 13 crosses truth with form for every policy, pooled over the families that ran each cell, Appendix I. The independence class writes the verbatim claim in 0.00 of configurations whether false or true, the reworted claim in 0.72 when false and 1.00 when true, and the independent one in 0.90 and 1.00. The governance rule writes no verbatim or reworted claim of either truth and the independent one in 0.95 and 1.00, since that source arrives typed as a register document. Share-all and the vote write nearly every cell. Two pairs move with truth: the independence class’s reworted pair, whose false half pools four families and whose true half the two that ran the cross set, 0.90 against 1.00 on Qwen and 0.60 against 1.00 on the closed family, and the closed-family judge’s independent pair, 0.60 against 0.70, after it refuses every verbatim pair.

Two diagnostic arms bound the collapse class. The lineage oracle, the independence vote with the authored copy relation in place of surface similarity, writes no verbatim or reworted claim of either truth and admits the independent one in 0.95 when false and 0.90 when true. Semantic collapse, the same vote with copy groups from bidirectional entailment under the pinned scorer, lets 0.10 of the reworted false claims and 0.15 of the reworted true ones through and admits the independent pair as the oracle does. Perfect copy detection removes the rewording failure of F5 but cannot establish the truth of independent support. The source type that places governance in F1 also admits the false restatement here.

## 5.7 The write-back loop

**F7. Switching off the agents’ retrieval of shared memory removes three quarters of what the closed family’s agents say and every declared dependency, and at most one false write.**

Share-all ran again with the agents’ retrieval of shared memory switched off, the policy still seeing the store and the consumer still reading it, on the 60 base configurations of Qwen and the closed family, Table 14 in

Appendix J. On the closed family candidates per configuration fall from 8.3 to 2.2 and candidates declaring a shared belief from 189 to 0, so three quarters of what the agents said followed a read of the store, while false candidates per configuration stay at 0.5, false writes at 28 and 27, configurations holding a false belief at 0.42 and 0.45, damage at 115 and 122, and adoption at 0.27 and 0.23. On Qwen, whose agents never declared a dependency, candidates per configuration go from 1.3 to 1.7 while false candidates stay at 0.6, false writes at 33 and damage at 147, with adoption 0.30 and 0.25. Under share-all, the loop adds agreement to falsehoods already supplied by the feeds while exposure changes little. Whether this agreement increases false admission under the policies of F1 that count support remains untested.

## 6 Conclusion

CPB fixes the lineage of every input source before a policy runs and observes the reach of every admitted belief after it. On it, agreement, consistency and source authority each leave a route by which a falsehood enters shared memory and reaches the consumer’s answer, admission follows the form of the second source more than the truth of the claim, a perfect copy relation included, and under share-all, switching the agents’ write-back off removes most of what the closed family’s agents say and at most one false write. Three limits, set out in Appendix S, bound the claims: parent independence in the correlated-agreement family is presumed from distinct citations, collapse is shown to refuse copies and not to admit corroboration, and the protective effect of a competing truth is measured in one family.

## References

- Anthropic. System card: Claude Opus 5. <https://www.anthropic.com/transparency>, July 2026.
- Jonas Becker, Jan Philip Wahle, Terry Ruas, and Bela Gipp. Misinformation propagation in benign multi-agent systems. *arXiv preprint arXiv:2606.16710*, 2026.
- Varun Pratap Bhardwaj, Garima Singh, and Arun Pratap Bhardwaj. SuperLocalMemory 4.0: The governed memory operating system for AI agents. *arXiv preprint arXiv:2608.08253*, 2026.
- Zhaorun Chen, Zhen Xiang, Chaowei Xiao, Dawn Song, and Bo Li. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases. *arXiv preprint arXiv:2407.12784*, 2024.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- Diego F. Cuadros, Abdoul-Aziz Maiga, Helen Meskhidze, and Andre Curtis-Trudel. Governed collaborative memory as artificial selection in LLM-based multi-agent systems. *arXiv preprint arXiv:2605.04264*, 2026.
- Tianyu Ding, Aditya Nannapaneni, Bingfan Liu, and Ling Zhang. Always-on agents: A survey of persistent memory, state, and governance in LLM agents. *arXiv preprint arXiv:2606.30306*, 2026.
- Shen Dong, Shaochen Xu, Pengfei He, Yige Li, Jiliang Tang, Tianming Liu, Hui Liu, and Zhen Xiang. Memory injection attacks on LLM agents via query-only interaction. *arXiv preprint arXiv:2503.03704*, 2025.

- Xin Luna Dong, Laure Berti-Equille, and Divesh Srivastava. Truth discovery and copying detection in a dynamic world. *Proceedings of the VLDB Endowment*, 2(1):562–573, 2009a. doi: 10.14778/1687627.1687691.
- Xin Luna Dong, Laure Berti-Equille, and Divesh Srivastava. Integrating conflicting data: the role of source dependence. *Proceedings of the VLDB Endowment*, 2(1):550–561, 2009b. doi: 10.14778/1687627.1687690.
- Xin Luna Dong, Laure Berti-Equille, Yifan Hu, and Divesh Srivastava. Global detection of complex copying relationships between sources. *Proceedings of the VLDB Endowment*, 3(1-2):1358–1369, 2010. doi: 10.14778/1920841.1921008.
- Joseph Fiorese, Parth Parag Kulkarni, Ashmal Vayani, Song Wang, and Mubarak Shah. Learning to share: Selective memory for efficient parallel agentic systems. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.05965.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Yupeng Han, Shuochen Liu, Kai Zhang, Ze Liu, Zhihong Pan, and Xianquan Wang. Learning what to share and what to personalize: Hierarchical strategy co-evolution for agent memory. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026. arXiv:2608.25329.
- Tianyu Hu, Weikai Lin, Weizhi Zhang, Jing Ma, and Song Wang. MemRouter: Memory-as-embedding routing for long-term conversational agents. *arXiv preprint arXiv:2605.00356*, 2026.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in LLM agents via incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257*, 2025. MemoryAgentBench.
- Laxmipriya Ganesh Iyer and Rahul Suresh Babu. Capability minimization as a safety primitive: Risk-aware causal gating for least-privilege LLM agents. *arXiv preprint arXiv:2606.13884*, 2026.
- Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- Guneet Kohli. Nine judges, two effective votes: Correlated errors undermine LLM evaluation panels. *arXiv preprint arXiv:2605.29800*, 2026.
- Weitong Kong, Di Wen, Kunyu Peng, David Schneider, Zeyun Zhong, Alexander Jaus, Zdravko Marinov, Jiale Wei, Ruiping Liu, Junwei Zheng, Yufan Chen, Lei Qi, and Rainer Stiefelwagen. IMPACT-CYCLE: A contract-based multi-agent system for claim-level supervisory correction of long-video semantic memory. *arXiv preprint arXiv:2604.20136*, 2026.
- Chris Latimer, Nicol   Boschi, Andrew Neeser, Chris Bartholomew, Gaurav Srivastava, Xuan Wang, and Naren Ramakrishnan. Hindsight is 20/20: Building agent memory that retains, recalls, and reflects. *arXiv preprint arXiv:2512.12818*, 2025.
- Baichuan Li, Junyi Yao, and Zihao Zheng. Remember, verify, or ask? cross-family evaluation of memory commitment in LLM agents. *arXiv preprint arXiv:2608.19564*, 2026a.
- Xiaoyang Li, Yiqi Wang, Haohui Lu, Zhi Chen, Mo Li, Pingan Song, Mingkai Zheng, and Taotao Cai. MemTX: Transactional belief commit for stateful agent memory. *arXiv preprint arXiv:2607.23929*, 2026b.

- Junfeng Liao, Qizhou Wang, Jianing Zhu, Bo Du, Rui Yan, and Xiuying Chen. Belief memory: Agent memory under partial observability. *arXiv preprint arXiv:2605.05583*, 2026.
- Chenchen Lin, Wenhao Yuan, Xuehe Wang, and Edith Cheuk Han Ngai. Beyond memory majority: Latent-source reasoning for multi-agent memory arbitration. *arXiv preprint arXiv:2608.19701*, 2026.
- Xun Liu, Yi Evie Zhang, Vira Kasprova, Parisa Rabbani, Pardis Sadat Zahraei, Tianyu Zhang, Ali Ebrahimpour-Boroojeny, and Varun Chandrasekaran. AgentAbstain: Do LLM agents know when not to act? *arXiv preprint arXiv:2607.10059*, 2026.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2402.17753.
- Yanki Margalit, Nurit Cohen-Inger, Erni Avram, Ran Taig, and Oded Margalit. Governed shared memory for multi-agent LLM systems. *arXiv preprint arXiv:2606.24535*, 2026.
- Kazuki Nakayashiki. When stale constraints go unchecked: Budgeted verification failures in inherited agent memory. *arXiv preprint arXiv:2608.25553*, 2026.
- Vassilis Papadopoulos, McNair Shah, Sam Zimmerman, and Jack Lindsey. Mind viruses: Self-propagating ideas in multi-agent LLM systems. *arXiv preprint arXiv:2608.10218*, 2026.
- Young Bin Park. Graph-native cognitive memory for AI agents: Formal belief revision semantics for versioned memory architectures. *arXiv preprint arXiv:2603.17244*, 2026.
- Fabio Petroni, Samuel Broscheit, Aleksandra Piktus, Patrick Lewis, Gautier Izacard, Lucas Hosseini, Jane Dwivedi-Yu, Maria Lomeli, Timo Schick, Pierre-Emmanuel Mazaré, Armand Joulin, Edouard Grave, and Sebastian Riedel. Improving Wikipedia verifiability with AI. *arXiv preprint arXiv:2207.06220*, 2022. WAFER / SIDE.
- Ravali Pochampally, Anish Das Sarma, Xin Luna Dong, Alexandra Meliou, and Divesh Srivastava. Fusing data with correlations. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, pages 433–444, 2014. doi: 10.1145/2588555.2593674.
- Yijiashun Qi, Xiang Xu, and Yuxuan Li. When not to write memory: Governing false promotion from correlated agent traces. In *MLISE 2026 Workshop*, 2026. arXiv:2607.02579.
- Qwen Team. Qwen3.5: Towards native multimodal agents. <https://qwen.ai/blog?id=qwen3.5>, February 2026.
- Md Hafizur Rahman, Zafaryab Haider, Tanzim Mahfuz, and Prabuddha Chakraborty. HARP: Measuring harm amplification in multi-agent LLM systems. *arXiv preprint arXiv:2605.27489*, 2026.
- Zhe Ren, Yibo Yang, Yimeng Chen, Zijun Zhao, Benshuo Fu, Zhihao Shu, Bingjie Zhang, Yangyang Xu, Dandan Guo, and Shuicheng Yan. GateMem: Benchmarking memory governance in multi-principal shared-memory agents. *arXiv preprint arXiv:2606.18829*, 2026.
- Alireza Rezazadeh, Zichao Li, Ange Lou, Yuying Zhao, Wei Wei, and Yujia Bao. Collaborative memory: Multi-user memory sharing in LLM agents with dynamic access control. *arXiv preprint arXiv:2505.18279*, 2025.

- Michael Schlichtkrull, Zhijiang Guo, and Andreas Vlachos. AVeriTeC: A dataset for real-world claim verification with evidence from the web. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023. arXiv:2305.13117.
- Bronislav Sidik and Lior Rokach. MEMTIER: Tiered memory architecture and retrieval bottleneck analysis for long-running autonomous AI agents. *arXiv preprint arXiv:2605.03675*, 2026.
- Pranav Singh. When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense. *arXiv preprint arXiv:2606.22030*, 2026.
- Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. ConflictBank: A benchmark for evaluating the influence of knowledge conflicts in LLM. *arXiv preprint arXiv:2408.12076*, 2024.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2018. arXiv:1803.05355.
- Yiqi Wang, Jiaqi Zhang, Taotao Cai, Zirui Liu, Qingqiang Sun, Zequn Sun, Zhangkai Wu, Manqing Dong, Mingkai Zheng, Xuefei Yin, and Yanming Zhu. From agent traces to trust: A survey of evidence tracing and execution provenance in LLM agents. *arXiv preprint arXiv:2606.04990*, 2026.
- Qianshan Wei, Tengchao Yang, Yaochen Wang, Xinfeng Li, Lijun Li, Zhenfei Yin, Yi Zhan, Thorsten Holz, Zhiqiang Lin, and XiaoFeng Wang. A-MemGuard: A proactive defense framework for LLM-based agent memory. *arXiv preprint arXiv:2510.02373*, 2025.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.10813.
- Shanglin Wu, Yuyang Luo, Yueqing Liang, Kaiwen Shi, Yanfang Ye, Ali Payani, and Kai Shu. Scaling teams or scaling time? memory enabled lifelong learning in LLM multi-agent systems. *arXiv preprint arXiv:2604.03295*, 2026.
- Zhishang Xiang, Zerui Chen, Yunbo Tang, Zhimin Wei, Ruqin Ning, Yujie Lin, Qinggang Zhang, and Jinsong Su. MemSyco-Bench: Benchmarking sycophancy in agent memory. *arXiv preprint arXiv:2607.01071*, 2026.
- Han Xiao, Hongjun Xu, Xin Zhang, Yidong Chen, and Xiaodong Shi. TARL: Transaction-aware reliable ledgers for executable memory management in long-term agents. *arXiv preprint arXiv:2608.03699*, 2026.
- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *International Conference on Learning Representations (ICLR)*, 2024. ConflictQA; arXiv:2305.13300.
- Naen Xu, Hengyu An, Shuo Shi, Jinghuai Zhang, Chunyi Zhou, Changjiang Li, Tianyu Du, Zhihui Fu, Jun Wang, and Shouling Ji. When agents “misremember” collectively: Exploring the Mandela effect in LLM-based multi-agent systems. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2602.00428.
- Tianyu Yang, Sudipta Paul, Vijay Srinivasan, Vivek Kulkarni, and Srinivas Chappidi. TRUSTMEM: Learning trustworthy memory consolidation for LLM agents with long-term memory. *arXiv preprint arXiv:2606.25161*, 2026.

- Guibin Zhang, Muxin Fu, Guancheng Wan, Miao Yu, Kun Wang, and Shuicheng Yan. G-Memory: Tracing hierarchical memory for multi-agent systems. *arXiv preprint arXiv:2506.07398*, 2025.
- Guilin Zhang, Wei Jiang, Xiejiashan Wang, Aisha Behr, Kai Zhao, Jeffrey Friedman, Xu Chu, and Amine Anoun. Adaptive memory admission control for LLM agents. *arXiv preprint arXiv:2603.04549*, 2026.
- Yan Zhang and Shibo Li. ConsistencyGate: Preventing memory contamination in LLM agents via self-consistency admission control. *arXiv preprint arXiv:2607.22962*, 2026.
- Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.14795.

## Organisation of the appendix

The appendix provides construction and policy details, supplementary results, and validation of the assertion grader. Sections follow the main text’s order, then the validation of the two graders, the judge material, the reproduction materials and the limitations. Tables are generated by committed scripts from stored results.

### A Coverage

Table 2 lists evaluation coverage. Confidence and unanimity baselines require inputs supplied by Static episodes but absent from individual Live candidates. The trained baselines also run only on Static. The additional frontier judge evaluates Qwen agents on 60 Live base configurations. Other admission policies run in both settings. Mem0 and A-MemGuard require a running store and are evaluated only on Live, and Table 4 gives their per-family coverage.

Table 2: Which policy ran on which half. Static is the frozen test split of 400 episodes, and Live is 100 committed configurations per agent family, composed as Section 4.2 states, plus the 30 stage-6 configurations on two families.

| Policy                                                                           | Static, 400 episodes | Live                                                  |
|----------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|
| Private only, share all, majority vote, governance rule                          | yes                  | four families, 100 configurations                     |
| Independence vote, DEPEND, review all correlated                                 | yes                  | four families, 100 configurations                     |
| Stage-6 cells of the cross, seven policies                                       | no                   | Qwen and Claude Opus 5, 30 configurations             |
| Lineage oracle, semantic collapse                                                | no                   | Qwen and Claude Opus 5, 70 dual-source configurations |
| Share all without the agents’ retrieval                                          | no                   | Qwen and Claude Opus 5, 60 base configurations        |
| Confidence threshold, unanimity vote                                             | yes                  | no                                                    |
| LLM judge                                                                        | four backbones       | each family by its own backbone                       |
| Frontier judge over Qwen agents                                                  | no                   | 60 base configurations                                |
| Trained baselines, logistic and boosted trees                                    | yes                  | no                                                    |
| Mem0 as the store under share all and under governance, A-MemGuard as the policy | no                   | four families, 100 configurations                     |

### B Corpus construction, audit and provenance ledger

**Adapters.** Claim-verification adapters supply insufficient-evidence verdicts for unverifiable and high-stakes scenarios, refuted claims with authoritative counter-evidence for misleading-majority scenarios, and supported claims with independent parents for the balance class. The citation graph supplies groups whose articles either cite one document or cite independent documents asserting the same proposition. Surface features are matched so that lineage distinguishes their gold actions. Evidence stance comes from source verdicts where available and a pinned natural language inference model otherwise. A pinned zero-shot classifier assigns each episode to one of four topical domains.

**Audit and freeze.** An audit applied four checks to 120 frozen test episodes and found two assembly defects. We corrected the code, rebuilt with unchanged seeds, audited the affected episodes and refroze the partition. The release reports residual stance errors and includes a field-level ledger of imported and authored content.

**Live configurations.** Each agent family runs the 100 configurations specified in Section 4.2. Configurations were committed before execution and are released with the store logs used for scoring. The scripts and the tables below number the sets as stages: stage 2 is the base set, stage 4 the dual-source set, stage 5 the reworted-copy set and stage 6 the cross set, and stages 1 and 3 were development sets that no reported number uses.

## C The policy view and the policies

**The view.** Static policies receive an evidence graph with inferred stances, surface-based copy groups, source-type authority proxies, timestamp-based freshness and observing agents. Live policies receive the candidate text, cited source texts and types, retrievable beliefs, and declared provenance. Live decisions operate on candidate evidence rather than a fixed set of agent votes. Policies never see the retrieval log, authored lineage or claim truth.

**Merged rows.** The independence vote, DEPEND and review-all-correlated rule make identical Live decisions on the three open families and share a table row. Their implementations use a common surface-collapse predicate, and declared provenance does not distinguish their decisions in these runs. They differ on the closed family, whose agents declare provenance on some candidates, and Table 3 gives their separate rows there.

Table 3: The three rules of the independence class where they separate. On the three open families they decide identically on every episode and share the independence row of Table 1. Columns as in that table.

| Policy                  | Claude Opus 5 |       |       |
|-------------------------|---------------|-------|-------|
|                         | Dmg↓          | FA↓   | TA↑   |
| Independent-source vote | 0.243         | 0.250 | 0.108 |
| DEPEND                  | 0.224         | 0.260 | 0.111 |
| Review all correlated   | 0.280         | 0.333 | 0.108 |

**The policies.** **B0** never promotes and **B1** always promotes. **B2** thresholds a confidence proxy, **B3** uses nominal majority support, and **B4** requires unanimity. **B5** requires two supporting groups after copy collapse. **B15**, DEPEND, estimates source accuracy and copying from statistical agreement and discounts suspected copiers (Dong et al., 2009b). Its comparison with B5 is pre-registered. **B16**, GovMem’s review-all-correlated rule, routes correlated support to review (Qi et al., 2026). **B7** promotes fresh authoritative support without contradiction, abstains under unresolved contradiction, and otherwise defers.

**B6** is an LLM judge receiving the same view as text. Two prompt variants were specified before tuning, and one was selected on validation and evaluated once per backbone on test. On Live, each family uses its own backbone as judge, with an additional closed-model judge over Qwen agents. Logistic regression and gradient-boosted trees use the same structured features and fit on the Static training split. Threshold tuning uses equal validation budgets across policies.

## D Two published systems inside the instrument

Table 4 compares two integrated components with share-all, governance and the local judge. Mem0 runs as the shared store using its shipped extraction prompt plus a domain instruction. Its extractor uses the family’s agent backbone through the same loaded instance, and its embedder is pinned. Admitted candidates enter its write path, and agents retrieve the resulting memories. We evaluate Mem0 under share-all and governance. A verbatim-storage control reproduces native-store decisions on the 60 Qwen base configurations.

A-MemGuard’s unchanged repository consistency checker receives the candidate, its cited sources and the five nearest visible beliefs under the same embedder. It promotes when all checks pass, keeps the candidate private otherwise, and requests evidence when nothing can be checked.

Refuse is the proportion of false candidates rejected. Hold is the proportion of configurations with a false belief at probe time. Adopt and Corr follow Appendix E. Absorb is the proportion of Mem0 admissions merged into existing memories.

Table 4: Mem0 and A-MemGuard compared with reference policies by agent family. **Ep.:** configurations. **Refuse:** proportion of false candidates rejected. **Hold:** proportion of configurations with false memory at probe time. **Adopt:** false-answer rate. **Corr:** true-answer rate among scenarios with a true value. **Absorb:** proportion of Mem0 admissions merged into existing memories. Mem0 replaces the store under share-all or governance, and A-MemGuard supplies the admission check. Both use the family’s backbone.

| Policy                          | Ep. | Refuse | Hold | Adopt | Corr | Absorb |
|---------------------------------|-----|--------|------|-------|------|--------|
| <i>Qwen3.5-9B</i>               |     |        |      |       |      |        |
| Share all                       | 100 | 0.00   | 0.62 | 0.45  | 0.65 | –      |
| Mem0 default write path         | 100 | 0.00   | 0.62 | 0.51  | 0.63 | 0.05   |
| Governance rule                 | 100 | 0.84   | 0.10 | 0.09  | 0.63 | –      |
| Mem0 behind the governance rule | 100 | 0.84   | 0.10 | 0.09  | 0.63 | 0.00   |
| A-MemGuard consistency check    | 100 | 0.00   | 0.62 | 0.45  | 0.63 | –      |
| LLM judge                       | 100 | 0.00   | 0.62 | 0.45  | 0.65 | –      |
| <i>Gemma-3-12B</i>              |     |        |      |       |      |        |
| Share all                       | 100 | 0.00   | 0.62 | 0.32  | 0.77 | –      |
| Mem0 default write path         | 100 | 0.00   | 0.62 | 0.32  | 0.68 | 0.67   |
| Governance rule                 | 100 | 0.85   | 0.09 | 0.06  | 0.58 | –      |
| Mem0 behind the governance rule | 100 | 0.92   | 0.09 | 0.06  | 0.58 | 0.00   |
| A-MemGuard consistency check    | 100 | 0.01   | 0.61 | 0.33  | 0.67 | –      |
| LLM judge                       | 100 | 0.01   | 0.61 | 0.33  | 0.68 | –      |
| <i>Llama-3.1-8B</i>             |     |        |      |       |      |        |
| Share all                       | 100 | 0.00   | 0.63 | 0.47  | 0.68 | –      |
| Mem0 default write path         | 100 | 0.00   | 0.60 | 0.41  | 0.67 | 0.86   |
| Governance rule                 | 100 | 0.86   | 0.10 | 0.07  | 0.65 | –      |
| Mem0 behind the governance rule | 100 | 0.92   | 0.10 | 0.07  | 0.63 | 0.04   |
| A-MemGuard consistency check    | 100 | 0.18   | 0.61 | 0.48  | 0.67 | –      |
| LLM judge                       | 100 | 0.68   | 0.30 | 0.22  | 0.78 | –      |
| <i>Claude Opus 5</i>            |     |        |      |       |      |        |
| Share all                       | 100 | 0.00   | 0.50 | 0.44  | 0.73 | –      |
| Mem0 default write path         | 100 | 0.00   | 0.50 | 0.43  | 0.75 | 0.39   |
| Governance rule                 | 100 | 0.88   | 0.06 | 0.06  | 0.60 | –      |
| Mem0 behind the governance rule | 100 | 0.88   | 0.06 | 0.07  | 0.52 | 0.05   |
| A-MemGuard consistency check    | 100 | 0.00   | 0.50 | 0.45  | 0.62 | –      |

## E Downstream adoption per policy and per family

The consumer answers one targeted question from shared memory at the final read, replayed from the committed transcript using the family’s backbone. Table 5 reports adoption, the proportion of probes asserting the false value, and correctness, the proportion asserting the true value among scenarios containing one. Empty stores yield abstention and zero on both measures. Figure 2 plots these rates.

Table 5: Downstream adoption on CPB-Live. The nine policies pool the 100 configurations of each family, the two diagnostic arms the 40 dual-source configurations of stages 4 and 5, and the frontier judge its 60 base configurations on Qwen. **Adopt**: the share of configurations in which the consumer asserted a false value when answering from shared memory. **Corr**: the share of configurations whose scenario carries a true value in which the consumer asserted it. The consumer runs on the family’s own backbone and receives no source of its own, so shared memory is its only information. Where a policy promotes nothing, both columns are zero by construction.

| Policy                   | Qwen3.5-9B |       | Gemma-3-12B |       | Llama-3.1-8B |       | Claude Opus 5 |       |
|--------------------------|------------|-------|-------------|-------|--------------|-------|---------------|-------|
|                          | Adopt↓     | Corr↑ | Adopt↓      | Corr↑ | Adopt↓       | Corr↑ | Adopt↓        | Corr↑ |
| Private only             | 0.00       | 0.00  | 0.00        | 0.00  | 0.00         | 0.00  | 0.00          | 0.00  |
| Share all                | 0.45       | 0.65  | 0.32        | 0.77  | 0.47         | 0.68  | 0.44          | 0.73  |
| Majority vote            | 0.45       | 0.65  | 0.34        | 0.62  | 0.45         | 0.77  | 0.41          | 0.62  |
| LLM judge                | 0.45       | 0.65  | 0.33        | 0.68  | 0.22         | 0.78  | 0.19          | 0.40  |
| Frontier judge over Qwen | 0.20       | 0.56  | –           | –     | –            | –     | –             | –     |
| Governance rule          | 0.09       | 0.63  | 0.06        | 0.58  | 0.07         | 0.65  | 0.06          | 0.60  |
| Independent-source vote  | 0.18       | 0.17  | 0.11        | 0.17  | 0.14         | 0.17  | 0.14          | 0.17  |
| DEPEN                    | 0.18       | 0.17  | 0.11        | 0.17  | 0.14         | 0.17  | 0.12          | 0.15  |
| Review all correlated    | 0.18       | 0.17  | 0.11        | 0.17  | 0.14         | 0.17  | 0.15          | 0.13  |
| Lineage oracle           | 0.00       | 1.00  | –           | –     | –            | –     | 0.00          | 0.80  |
| Semantic collapse        | 0.07       | 1.00  | –           | –     | –            | –     | 0.05          | 0.90  |

Table 6 repeats both columns under the robustness matcher of Appendix M, which writes numeric forms and articles in one canonical way before matching, so that one-fifth and 20 percent match a fifth and 50 percent matches half. On the human sample it reaches agreement 0.975 with the first author and recall 1.000 on the false label. Across the four families it moves adoption in no cell by more than three hundredths and correctness by at most five hundredths, and leaves the ordering of the policies unchanged, so the misses of the rule of record fall evenly across policies.

Table 6: Downstream adoption under the rule of record and under the robustness matcher, on the pools of Table 5, whose **Adopt** and **Corr** columns repeat here. **norm** marks the same quantity after answer and gold value are mapped to a canonical form by a fixed table, so that a fraction, its percentage and its hyphenated spelling count as one value and a leading article is ignored. Every difference between a plain column and its norm column is an answer the rule of record graded as other because the value was stated in another form.

| Policy                   | Qwen3.5-9B |      |      |      | Gemma-3-12B |      |      |      | Llama-3.1-8B |      |      |      | Claude Opus 5 |      |      |      |
|--------------------------|------------|------|------|------|-------------|------|------|------|--------------|------|------|------|---------------|------|------|------|
|                          | Adopt      | norm | Corr | norm | Adopt       | norm | Corr | norm | Adopt        | norm | Corr | norm | Adopt         | norm | Corr | norm |
| Private only             | 0.00       | 0.00 | 0.00 | 0.00 | 0.00        | 0.00 | 0.00 | 0.00 | 0.00         | 0.00 | 0.00 | 0.00 | 0.00          | 0.00 | 0.00 | 0.00 |
| Share all                | 0.45       | 0.47 | 0.65 | 0.67 | 0.32        | 0.32 | 0.77 | 0.78 | 0.47         | 0.47 | 0.68 | 0.68 | 0.44          | 0.43 | 0.73 | 0.77 |
| Majority vote            | 0.45       | 0.47 | 0.65 | 0.67 | 0.34        | 0.35 | 0.62 | 0.63 | 0.45         | 0.45 | 0.77 | 0.77 | 0.41          | 0.41 | 0.62 | 0.63 |
| LLM judge                | 0.45       | 0.47 | 0.65 | 0.67 | 0.33        | 0.33 | 0.68 | 0.70 | 0.22         | 0.22 | 0.78 | 0.78 | 0.19          | 0.21 | 0.40 | 0.45 |
| Frontier judge over Qwen | 0.20       | 0.20 | 0.56 | 0.58 | –           | –    | –    | –    | –            | –    | –    | –    | –             | –    | –    | –    |
| Governance rule          | 0.09       | 0.10 | 0.63 | 0.63 | 0.06        | 0.06 | 0.58 | 0.58 | 0.07         | 0.07 | 0.65 | 0.65 | 0.06          | 0.06 | 0.60 | 0.60 |
| Independent-source vote  | 0.18       | 0.19 | 0.17 | 0.17 | 0.11        | 0.11 | 0.17 | 0.17 | 0.14         | 0.14 | 0.17 | 0.17 | 0.14          | 0.14 | 0.17 | 0.17 |
| DEPEN                    | 0.18       | 0.19 | 0.17 | 0.17 | 0.11        | 0.11 | 0.17 | 0.17 | 0.14         | 0.14 | 0.17 | 0.17 | 0.12          | 0.12 | 0.15 | 0.15 |
| Review all correlated    | 0.18       | 0.19 | 0.17 | 0.17 | 0.11        | 0.11 | 0.17 | 0.17 | 0.14         | 0.14 | 0.17 | 0.17 | 0.15          | 0.15 | 0.13 | 0.13 |
| Lineage oracle           | 0.00       | 0.00 | 1.00 | 1.00 | –           | –    | –    | –    | –            | –    | –    | –    | 0.00          | 0.00 | 0.80 | 0.80 |
| Semantic collapse        | 0.07       | 0.10 | 1.00 | 1.00 | –           | –    | –    | –    | –            | –    | –    | –    | 0.05          | 0.05 | 0.90 | 0.90 |

## F Adoption as a function of what shared memory held

Table 7 groups probes by the number of visible false beliefs and the presence of a true belief, pooled over policies and agent families. Empty and true-only stores provide reference conditions. The one-false-belief groups support the comparison in Section 5.3. Groups with multiple false beliefs are small and support only descriptive observations.

Table 7: Adoption as a function of what shared memory holds when the consumer is asked, pooled over policies and agent families. A competing true belief is strongly protective, and the protection weakens as false beliefs accumulate.

| Visible false beliefs | True belief also visible | Probes | Asserted the false value |
|-----------------------|--------------------------|--------|--------------------------|
| 0                     | no                       | 1729   | 0.01                     |
| 0                     | yes                      | 707    | 0.00                     |
| 1                     | no                       | 553    | 0.98                     |
| 1                     | yes                      | 160    | 0.15                     |
| 2                     | no                       | 22     | 0.95                     |
| 2                     | yes                      | 8      | 0.00                     |
| 3+                    | no                       | 57     | 0.89                     |
| 3+                    | yes                      | 24     | 0.42                     |

## G Refusal conditioned on a false candidate arriving

Table 8 reports the proportion of false candidates rejected, with denominators in brackets. Truth follows the assertion grade in Appendix L. Normalising by candidates reduces differences in output volume, but each

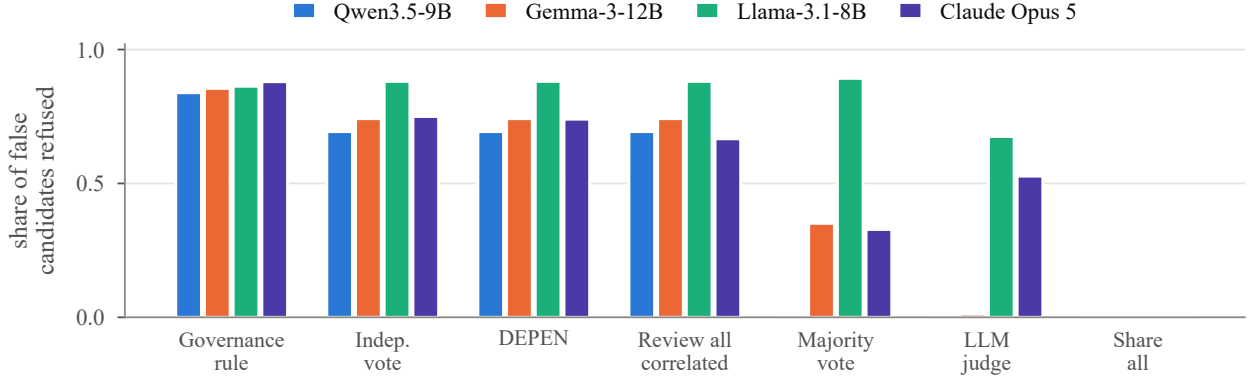


Figure 5: Refusal among false candidates on each policy’s own stream, the eight policies on the 100 configurations of each family. Bars distinguish agent families, and Table 8 gives the denominators and the rows of the diagnostic arms and the frontier judge over Qwen on their own pools.

policy still faces its own stream because earlier admissions affect later claims. These rates therefore describe decisions on encountered candidates, not a comparison on fixed inputs. Private only and share all define the endpoints.

Table 8: Refusal conditioned on a false candidate arriving: of the false candidates a policy was asked to decide, false by the assertion grade, the share it did not promote. The denominator is in brackets and is each policy’s own stream over the pool of Table 5. Conditioning removes the families’ difference in verbosity and not the difference between the streams the policies faced. Private only and Share all bound the range by construction.

| Policy                   | Qwen3.5-9B | Gemma-3-12B | Llama-3.1-8B | Claude Opus 5 |
|--------------------------|------------|-------------|--------------|---------------|
| Private only             | 1.00 [62]  | 1.00 [62]   | 1.00 [67]    | 1.00 [48]     |
| Share all                | 0.00 [62]  | 0.00 [81]   | 0.00 [735]   | 0.00 [56]     |
| Majority vote            | 0.00 [62]  | 0.35 [94]   | 0.89 [604]   | 0.33 [64]     |
| LLM judge                | 0.00 [62]  | 0.01 [81]   | 0.68 [345]   | 0.53 [55]     |
| Frontier judge over Qwen | 0.51 [35]  | –           | –            | –             |
| Governance rule          | 0.84 [62]  | 0.85 [62]   | 0.86 [73]    | 0.88 [50]     |
| Independent-source vote  | 0.69 [62]  | 0.74 [62]   | 0.88 [143]   | 0.75 [52]     |
| DEPEND                   | 0.69 [62]  | 0.74 [62]   | 0.88 [143]   | 0.74 [50]     |
| Review all correlated    | 0.69 [62]  | 0.74 [62]   | 0.88 [143]   | 0.67 [48]     |
| Lineage oracle           | 1.00 [29]  | –           | –            | 1.00 [25]     |
| Semantic collapse        | 0.86 [29]  | –           | –            | 0.92 [24]     |

## H Per-class breakdown of the Live results

The following tables break down Table 1 by scenario class and agent family. Each cell reports damage, false writes and true writes. The first three columns cover the 60 base configurations: correlated agreement, concurrent conflict and mixed scenarios. COPY and IND contain verbatim false copies and independent true restatements. PARA and PARA-AUTH contain reworded false copies with web and authoritative source types. TCOPY, TPARA and FIND are the stage-6 cells of Table 13, run on Qwen and the closed family, and

lie outside the pool of Table 1. Row sums over the first seven columns reproduce the pooled damage and write totals underlying Table 1.

Table 9: Qwen3.5-9B: damage, false writes and true writes per scenario class, the totals of Table 1 attributed to the classes that produced them. I-CORR, D-CONC and mixed are the 60 base configurations. COPY is the dual-source set with a verbatim copy of a false claim and IND the one with an independent restatement of a true claim. PARA is a reworded copy of a false claim as web text and PARA-AUTH the same copy typed authoritative. TCOPY and TPARA are a verbatim and a reworded copy of a true claim and FIND an independent restatement of a false claim. These three classes are outside the pool of Table 1.

| Policy                  | I-CORR   | D-CONC   | mixed   | COPY    | IND    | PARA   | PARA-AUTH | TCOPY  | TPARA  | FIND    |
|-------------------------|----------|----------|---------|---------|--------|--------|-----------|--------|--------|---------|
| Private only            | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0   | 0·0·0  | 0·0·0  | 0·0·0     | —      | —      | —       |
| Share all               | 50·10·10 | 72·18·20 | 25·5·15 | 40·10·0 | 0·0·10 | 36·9·0 | 40·10·0   | 0·0·10 | 0·0·10 | 40·10·0 |
| Majority vote           | 50·10·10 | 72·18·18 | 25·5·15 | 40·10·0 | 0·0·10 | 36·9·0 | 40·10·0   | 0·0·10 | 0·0·10 | 40·10·0 |
| LLM judge               | 50·10·10 | 72·18·20 | 25·5·15 | 40·10·0 | 0·0·10 | 36·9·0 | 40·10·0   | 0·0·10 | 0·0·10 | 40·10·0 |
| Governance rule         | 0·0·10   | 0·0·18   | 0·0·0   | 0·0·0   | 0·0·10 | 0·0·0  | 40·10·0   | 0·0·0  | 0·0·0  | 40·10·0 |
| Independent-source vote | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0   | 0·0·10 | 36·9·0 | 40·10·0   | 0·0·0  | 0·0·10 | 40·10·0 |
| DEPEN                   | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0   | 0·0·10 | 36·9·0 | 40·10·0   | —      | —      | —       |
| Review all correlated   | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0   | 0·0·10 | 36·9·0 | 40·10·0   | —      | —      | —       |
| Lineage oracle          | —        | —        | —       | —       | —      | —      | —         | 0·0·0  | 0·0·0  | 40·10·0 |
| Semantic collapse       | —        | —        | —       | —       | —      | —      | —         | 0·0·0  | 0·0·2  | 40·10·0 |

Table 10: Gemma-3-12B: damage, false writes and true writes per scenario class, the totals of Table 1 attributed to the classes that produced them. I-CORR, D-CONC and mixed are the 60 base configurations. COPY is the dual-source set with a verbatim copy of a false claim and IND the one with an independent restatement of a true claim. PARA is a reworded copy of a false claim as web text and PARA-AUTH the same copy typed authoritative.

| Policy                  | I-CORR   | D-CONC    | mixed   | COPY    | IND    | PARA    | PARA-AUTH |
|-------------------------|----------|-----------|---------|---------|--------|---------|-----------|
| Private only            | 0·0·0    | 0·0·0     | 0·0·0   | 0·0·0   | 0·0·0  | 0·0·0   | 0·0·0     |
| Share all               | 50·10·53 | 72·35·109 | 20·6·53 | 40·10·0 | 0·0·10 | 40·10·0 | 40·10·0   |
| Majority vote           | 50·10·15 | 72·18·22  | 20·4·15 | 40·10·0 | 0·0·10 | 36·9·0  | 40·10·0   |
| LLM judge               | 50·10·53 | 72·35·80  | 20·6·53 | 40·10·0 | 0·0·10 | 36·9·0  | 40·10·0   |
| Governance rule         | 0·0·10   | 0·0·18    | 0·0·0   | 0·0·0   | 0·0·10 | 0·0·0   | 36·9·0    |
| Independent-source vote | 0·0·0    | 0·0·0     | 0·0·0   | 0·0·0   | 0·0·10 | 28·7·0  | 36·9·0    |
| DEPEN                   | 0·0·0    | 0·0·0     | 0·0·0   | 0·0·0   | 0·0·10 | 28·7·0  | 36·9·0    |
| Review all correlated   | 0·0·0    | 0·0·0     | 0·0·0   | 0·0·0   | 0·0·10 | 28·7·0  | 36·9·0    |

Table 11: Llama-3.1-8B: damage, false writes and true writes per scenario class, the totals of Table 1 attributed to the classes that produced them. I-CORR, D-CONC and mixed are the 60 base configurations. COPY is the dual-source set with a verbatim copy of a false claim and IND the one with an independent restatement of a true claim. PARA is a reworded copy of a false claim as web text and PARA-AUTH the same copy typed authoritative.

| Policy                  | I-CORR    | D-CONC     | mixed     | COPY     | IND     | PARA     | PARA-AUTH |
|-------------------------|-----------|------------|-----------|----------|---------|----------|-----------|
| Private only            | 0·0·0     | 0·0·0      | 0·0·0     | 0·0·0    | 0·0·0   | 0·0·0    | 0·0·0     |
| Share all               | 40·61·161 | 80·170·231 | 25·76·204 | 40·132·0 | 0·0·143 | 40·182·0 | 40·114·0  |
| Majority vote           | 35·7·19   | 80·20·23   | 25·5·14   | 40·10·0  | 0·0·10  | 40·12·0  | 40·11·0   |
| LLM judge               | 20·5·82   | 32·26·36   | 20·17·42  | 8·15·0   | 0·0·22  | 16·25·0  | 32·24·0   |
| Governance rule         | 0·0·10    | 0·0·20     | 0·0·0     | 0·0·0    | 0·0·10  | 0·0·0    | 40·10·0   |
| Independent-source vote | 0·0·0     | 0·0·0      | 0·0·0     | 0·0·0    | 0·0·10  | 28·7·0   | 40·10·0   |
| DEPEN                   | 0·0·0     | 0·0·0      | 0·0·0     | 0·0·0    | 0·0·10  | 28·7·0   | 40·10·0   |
| Review all correlated   | 0·0·0     | 0·0·0      | 0·0·0     | 0·0·0    | 0·0·10  | 28·7·0   | 40·10·0   |

Table 12: Claude Opus 5: damage, false writes and true writes per scenario class, the totals of Table 1 attributed to the classes that produced them. I-CORR, D-CONC and mixed are the 60 base configurations. COPY is the dual-source set with a verbatim copy of a false claim and IND the one with an independent restatement of a true claim. PARA is a reworded copy of a false claim as web text and PARA-AUTH the same copy typed authoritative. TCOPY and TPARA are a verbatim and a reworded copy of a true claim and FIND an independent restatement of a false claim. These three classes are outside the pool of Table 1.

| Policy                  | I-CORR   | D-CONC   | mixed   | COPY   | IND    | PARA   | PARA-AUTH | TCOPY  | TPARA  | FIND    |
|-------------------------|----------|----------|---------|--------|--------|--------|-----------|--------|--------|---------|
| Private only            | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0  | 0·0·0  | 0·0·0  | 0·0·0     | —      | —      | —       |
| Share all               | 50·10·19 | 40·12·79 | 25·6·21 | 24·6·0 | 0·0·25 | 36·9·0 | 39·13·0   | 0·0·12 | 0·0·10 | 40·14·0 |
| Majority vote           | 45·9·10  | 40·10·14 | 20·4·12 | 24·6·0 | 0·0·9  | 32·8·0 | 24·6·0    | 0·0·8  | 0·0·9  | 36·9·0  |
| LLM judge               | 35·9·12  | 44·12·28 | 15·3·14 | 0·0·0  | 0·0·10 | 4·1·0  | 3·1·0     | 0·0·0  | 0·0·1  | 23·8·0  |
| Governance rule         | 0·0·4    | 0·0·16   | 0·0·0   | 0·0·0  | 0·0·11 | 0·0·0  | 24·6·0    | 0·0·0  | 0·0·0  | 36·9·0  |
| Independent-source vote | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0  | 0·0·10 | 24·6·0 | 28·7·0    | 0·0·0  | 0·0·10 | 32·8·0  |
| DEPEN                   | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0  | 0·0·10 | 24·7·0 | 24·6·0    | —      | —      | —       |
| Review all correlated   | 0·0·0    | 0·0·0    | 0·0·0   | 0·0·0  | 0·0·9  | 32·9·0 | 28·7·0    | —      | —      | —       |
| Lineage oracle          | —        | —        | —       | —      | —      | —      | —         | 0·0·0  | 0·0·0  | 36·10·0 |
| Semantic collapse       | —        | —        | —       | —      | —      | —      | —         | 0·0·0  | 0·0·1  | 36·9·0  |

## I Truth against form

Table 13 places every policy on the six cells of the dual-source design, the false and the true claim under each of three forms of the second source: a verbatim copy, a reworded copy from the same lineage, and an independent restatement typed as a register document. The false cells are the stage-4 verbatim copy, the stage-5 reworded copy typed as web text and the stage-6 independent restatement, the true cells the stage-6 verbatim and reworded copies and the stage-4 independent restatement. A cell is the share of the class’s episodes in which the policy wrote the target proposition, the false one in a false cell and the true one in a true cell. The blocks are the two families that ran stage 6 and the pool over every family that ran a cell, so the pooled denominators differ between the stage-4 and stage-5 columns, which four families ran, and the stage-6 columns, which two ran. The lineage oracle and semantic collapse rows are the two diagnostic arms of Section 5.6. Their refusal and adoption on the stage-4 and stage-5 dual-source sets appear as their own rows of Tables 8 and 5.

Table 13: Writes of the target proposition on CPB-Live by the form of the second source and the truth of the claim. Each cell is the share of configurations of that class in which the policy wrote the proposition. A verbatim pair is a claim and a copy of itself under another source id, a reworded pair restates the claim in other words from the same lineage, and an independent pair adds a genuinely independent restatement. The false cells come from the COPY, PARA and FIND classes and the true cells from the TCOPY, TPARA and IND classes. A lower false share and a higher true share are better. The judge row is each family’s own model. A missing cell is a class the family has not run.

| Policy                      | Verbatim |       | Reworded |       | Independent |       |
|-----------------------------|----------|-------|----------|-------|-------------|-------|
|                             | false↓   | true↑ | false↓   | true↑ | false↓      | true↑ |
| <i>Qwen3.5-9B</i>           |          |       |          |       |             |       |
| Private only                | 0.00     | –     | 0.00     | –     | –           | 0.00  |
| Share all                   | 1.00     | 1.00  | 0.90     | 1.00  | 1.00        | 1.00  |
| Majority vote               | 1.00     | 1.00  | 0.90     | 1.00  | 1.00        | 1.00  |
| LLM judge                   | 1.00     | 1.00  | 0.90     | 1.00  | 1.00        | 1.00  |
| Governance rule             | 0.00     | 0.00  | 0.00     | 0.00  | 1.00        | 1.00  |
| Independent-source vote     | 0.00     | 0.00  | 0.90     | 1.00  | 1.00        | 1.00  |
| DEPEN                       | 0.00     | –     | 0.90     | –     | –           | 1.00  |
| Review all correlated       | 0.00     | –     | 0.90     | –     | –           | 1.00  |
| Lineage oracle              | 0.00     | 0.00  | 0.00     | 0.00  | 1.00        | 1.00  |
| Semantic collapse           | 0.00     | 0.00  | 0.10     | 0.20  | 1.00        | 1.00  |
| <i>Claude Opus 5</i>        |          |       |          |       |             |       |
| Private only                | 0.00     | –     | 0.00     | –     | –           | 0.00  |
| Share all                   | 0.60     | 0.90  | 0.90     | 1.00  | 1.00        | 1.00  |
| Majority vote               | 0.60     | 0.80  | 0.80     | 0.90  | 0.90        | 0.90  |
| LLM judge                   | 0.00     | 0.00  | 0.10     | 0.10  | 0.60        | 0.70  |
| Governance rule             | 0.00     | 0.00  | 0.00     | 0.00  | 0.90        | 1.00  |
| Independent-source vote     | 0.00     | 0.00  | 0.60     | 1.00  | 0.80        | 1.00  |
| DEPEN                       | 0.00     | –     | 0.60     | –     | –           | 1.00  |
| Review all correlated       | 0.00     | –     | 0.80     | –     | –           | 0.90  |
| Lineage oracle              | 0.00     | 0.00  | 0.00     | 0.00  | 0.90        | 0.80  |
| Semantic collapse           | 0.00     | 0.00  | 0.10     | 0.10  | 0.90        | 0.90  |
| <i>Pooled over families</i> |          |       |          |       |             |       |
| Private only                | 0.00     | –     | 0.00     | –     | –           | 0.00  |
| Share all                   | 0.90     | 0.95  | 0.95     | 1.00  | 1.00        | 1.00  |
| Majority vote               | 0.90     | 0.90  | 0.90     | 0.95  | 0.95        | 0.97  |
| LLM judge                   | 0.55     | 0.50  | 0.57     | 0.55  | 0.80        | 0.93  |
| Governance rule             | 0.00     | 0.00  | 0.00     | 0.00  | 0.95        | 1.00  |
| Independent-source vote     | 0.00     | 0.00  | 0.72     | 1.00  | 0.90        | 1.00  |
| DEPEN                       | 0.00     | –     | 0.72     | –     | –           | 1.00  |
| Review all correlated       | 0.00     | –     | 0.78     | –     | –           | 0.97  |
| Lineage oracle              | 0.00     | 0.00  | 0.00     | 0.00  | 0.95        | 0.90  |
| Semantic collapse           | 0.00     | 0.00  | 0.10     | 0.15  | 0.95        | 0.95  |

Every family cell is a share over 10 episodes and the pooled block sums the families.

## J The write-back ablation

Table 14 gives share-all on the 60 base configurations twice, with the agents retrieving shared memory each round and with that retrieval switched off while the policy still sees the store and the consumer still reads it. Correlated agreement and the declared dependencies are zero by construction in the ablated arm and are listed to show it. Damage and adoption are computed exactly as in the main arm, from the consumer’s reads and the probe. Section 5.7 reads the table.

Table 14: The shared memory writeback ablation on the stage 2 configurations. Share all is run once with every emitting agent retrieving shared memory and once with that retrieval switched off while the consumers still read the store. **Ep.** counts configurations. **Cand.** is the number of candidates per configuration and **False cand.** the number of false candidates per configuration. **Writes** is the number of false beliefs written. **Hold** is the proportion of configurations holding a false belief. **Damage** is the number of consumer reads that touched a false belief. **Decl.** is the number of candidates that declared a shared belief as their basis. **Adopt** is the false answer rate at the probe. Families that did not run the ablated arm are not listed.

| Arm                            | Ep. | Cand. | False cand. | Writes | Hold | Damage | Decl. | Adopt |
|--------------------------------|-----|-------|-------------|--------|------|--------|-------|-------|
| <i>Qwen3.5-9B</i>              |     |       |             |        |      |        |       |       |
| Share all                      | 60  | 1.3   | 0.6         | 33     | 0.55 | 147    | 0     | 0.30  |
| Share all, no shared retrieval | 60  | 1.7   | 0.6         | 33     | 0.55 | 147    | 0     | 0.25  |
| <i>Claude Opus 5</i>           |     |       |             |        |      |        |       |       |
| Share all                      | 60  | 8.3   | 0.5         | 28     | 0.42 | 115    | 189   | 0.27  |
| Share all, no shared retrieval | 60  | 2.2   | 0.5         | 27     | 0.45 | 122    | 0     | 0.23  |

## K Correction, arrival order and declared provenance

The environment offers contest, demote and supersede each round, but no compared policy retracts, so none of the 1,695 admitted false beliefs is corrected, and the first false write lands in round one or two in all but two episodes. Over ten swapped-order pairs, sensitivity is 0.00 for every policy on Qwen, while under the same greedy decoding share-all flips on 0.6 of Gemma pairs and every Llama pair: arrival order decides admission wherever the agents’ output varies with it.

Agents were told to declare any shared belief a claim rests on. Qwen agents declared none on their 78 candidates on the base configurations, and the share of false writes with a later declared dependency runs from 0.01 on Qwen to 0.82 on Llama, so declaration-based amplification is a lower bound shaped by reporting behaviour.

## L How a live candidate is judged false

We considered three ways to label Live candidates. The original provenance trace uses exact text, cited false feeds or declared reliance on false beliefs. It can miss uncited restatements and count hedges citing false sources. The content rule detects a false filler without the true filler. Designed for ten-word probe answers, it can misclassify negations or mentions within other propositions.

The reported results use an assertion grade. Claude Opus 5, through the same API interface as the frontier judge, labels each distinct pair of scenario and candidate as asserting the false proposition, asserting the true proposition, or neither. One fixed prompt supplies the candidate and verbatim gold propositions. The *neither* label covers mentions, questions, hedges, negations, discussion and unrelated assertions. If the scenario has no false proposition, the prompt states this and no candidate is labelled false. All responses and labels are persisted. The grading covers 6,443 distinct pairs of scenario and sentence across four families, all policies and arms and both integrated components.

One author labelled the 120 pairs, 30 per family, from the text alone before seeing the grader’s labels: agreement 0.867 and Cohen’s kappa 0.791, with 16 disagreements running both ways, presuppositions and side facts the grader read as assertions and attributed reports and exclusions it read as neither. The author then reviewed those 16 with both labels visible and changed 9, 8 of them toward the grader, and Table 15 reports the reviewed labels: agreement 0.933 and Cohen’s kappa 0.896. The blind figure is the validation and the reviewed one is where the author settled.

On the reviewed labels the grader’s precision on the false label is 0.958 and its recall 0.852: of the 24 pairs it called false the author called 1 neither, a rate whose Wilson interval runs from 0.01 to 0.20, and of the 27 the author called false it called 4 neither, all reports attributed to a source or sentences excluding a named value. The false counts in the paper therefore lean low rather than high against the author’s reading, and the sample shows the direction of the errors and not their rate over the full set. Agreement is 30, 29, 27 and 26 of 30 on Qwen, Gemma, Llama and the closed family. The closed-family agents and grader are the same model, and three of its four disagreements are sentences the author called false and the grader called neither, and the fourth one the author called neither and the grader called true, so the pairing does not favour that family.

All reported false-write, reach, damage and refusal results use assertion grading. The original provenance-based scores are archived. Adoption retains the separate grading rule for short consumer answers.

Table 15: The assertion grade against two annotators on 120 pairs, 30 per family. Left: annotator A’s reviewed label by row and the grader’s by column. Middle: pairs on which the two agree, per family. Right: agreement and Cohen’s kappa for every pairing of labels, blind and reviewed; the blind rows are the independent validation. Of the pairs the grader called false, 1 of 24 are not assertions to the annotator, a rate whose Wilson interval runs from 0.01 to 0.20.

| A \ Grader    | asserts false | asserts true | neither | total | Family        | Pairs | Agree |
|---------------|---------------|--------------|---------|-------|---------------|-------|-------|
| asserts false | 23            | 0            | 4       | 27    | Qwen3.5-9B    | 30    | 30    |
| asserts true  | 0             | 42           | 2       | 44    | Gemma-3-12B   | 30    | 29    |
| neither       | 1             | 1            | 47      | 49    | Llama-3.1-8B  | 30    | 27    |
| total         | 24            | 43           | 53      | 120   | Claude Opus 5 | 30    | 26    |

| Labels              | Agree | $\kappa$ |
|---------------------|-------|----------|
| A blind, grader     | 0.867 | 0.791    |
| A reviewed, grader  | 0.933 | 0.896    |
| B blind, grader     | 0.875 | 0.806    |
| A blind, B blind    | 0.958 | 0.935    |
| A reviewed, B blind | 0.933 | 0.897    |

## M Human validation of the two graders

Two people label two blind samples from the text alone, with no grader or rule output shown. The first sample is the 120 pairs of Appendix L, labelled as asserting the false proposition, the true proposition or neither, whose first author’s labels Table 15 reports. The second is 120 consumer answers drawn from the four families, six per family and per category of the probe’s rule, labelled as asserting the false value, the true value, both, abstaining or something else. On the first author’s blind labels the rule agrees on 0.633 of answers, almost all of the shortfall being the 24 answers that consist of the word unknown, which the author had labelled by the value the store held rather than by the answer. After the author reviewed the 44 disagreements with the rule’s category visible, Table 16 reports agreement 0.842 and Cohen’s kappa 0.802. The rule’s false label has precision 1.000 and recall 0.706 against the author: every answer it calls false is one, and it misses 10 answers that state the false value in another form, a fifth written as one-fifth or 20 percent, half written as 50 percent, a name written without its article. Those answers fall in the rule’s other category, of which the author calls 10 false, 6 true, 1 an abstention and 7 other. Adoption and correctness in the paper are therefore undercounts wherever a consumer rephrases a value, and Appendix E reports both columns again under a matcher that normalises numbers and articles, which recovers every one of those misses on the sample. A second person labelled the same 120 answers blind and agrees with the first author’s reviewed labels on 0.958 of them, Cohen’s kappa 0.946, with the first author’s blind labels on 0.800, and with the rule on 0.808. All three agree on 97 answers. On the assertion sample the second person, labelling blind, agrees with the first author’s blind labels on 0.958 of pairs and with the reviewed ones on 0.933 with Cohen’s kappa 0.897, and with the grader on 0.875 with kappa 0.806. All three agree on 105 of the 120 pairs. The two people agree with each other more than either agrees with the grader, and the review moved the first author toward the grader and away from the second person, so the blind rows of Tables 15 and 16 are the ones to read.

Table 16: The consumer probe’s grading rule against two annotators on 120 answers, six per family and per category of the rule. Left: annotator A’s reviewed label by row and the rule’s by column. Middle: answers on which the two agree, per family. Right: agreement and Cohen’s kappa for every pairing of labels, blind and reviewed; the blind rows are the independent validation, and A’s blind shortfall is the 24 answers reading unknown that A had labelled by the stored value.

| A \ rule    | false value | true value | both | abstains | other | total |
|-------------|-------------|------------|------|----------|-------|-------|
| false value | 24          | 0          | 0    | 0        | 10    | 34    |
| true value  | 0           | 24         | 1    | 0        | 6     | 31    |
| both        | 0           | 0          | 22   | 0        | 0     | 22    |
| abstains    | 0           | 0          | 1    | 24       | 1     | 26    |
| other       | 0           | 0          | 0    | 0        | 7     | 7     |
| total       | 24          | 24         | 24   | 24       | 24    | 120   |

| Family        | Answers | Agree |
|---------------|---------|-------|
| Qwen3.5-9B    | 30      | 24    |
| Gemma-3-12B   | 30      | 24    |
| Llama-3.1-8B  | 30      | 30    |
| Claude Opus 5 | 30      | 23    |

| Labels              | Agree | $\kappa$ |
|---------------------|-------|----------|
| A blind, rule       | 0.633 | 0.542    |
| A reviewed, rule    | 0.842 | 0.802    |
| B blind, rule       | 0.808 | 0.760    |
| A blind, B blind    | 0.800 | 0.737    |
| A reviewed, B blind | 0.958 | 0.946    |

## N Assertion and provenance measure different properties

Every false write uses the assertion grader validated in Appendix L. Two earlier rules are measured against it: the provenance trace, which marks a candidate false by exact text, a cited false feed or a declared false belief, and the content rule of the consumer probe, a false filler present and no true one. Table 17 sets the three side by side under share-all.

Over the nine policies the trace agrees with the grade on every Qwen candidate. On Llama it misses 1,524 false assertions and counts 120 candidates that assert nothing false. On the closed family it counts 839 such candidates, hedges and meta statements citing false feeds, beside 405 it gets right, and dropping them lowers share-all’s damage from 275 to 214, while damage moves by at most 13 on the other families under share-all because most newly counted restatements fall in episodes already polluted.

The two errors are distinct: a claim can repeat a falsehood without naming its source, and a citation can mention one without endorsing it. The grade therefore judges the sentence, and provenance keeps its role of tracing where a claim came from.

In Table 17 the top block gives false-candidate counts and damage under each rule, plus content-detected cases missed by provenance tracing. It also identifies missed cases with neither citations nor declared dependencies. The bottom block reports refusal under the historical rules, and current refusal appears in Figure 5. Content counts exclude scenarios whose gold the probe cannot parse, with exclusions reported by the generating script.

Table 17: Assertion grading versus historical scoring rules. Top: share-all false-candidate counts under Grade, Trace and Content, with damage under Grade, Trace and the union of Trace and Content, written Both. Missed denotes content-detected cases absent from the trace, Uncited those with neither citations nor declared dependencies, and Admitted those promoted. Bottom: historical refusal rates with denominators in brackets. Current refusal appears in Figure 5.

| Family        | Grade | Trace | Content | Missed | Uncited | Admitted | Dmg grade | Dmg trace | Dmg both |
|---------------|-------|-------|---------|--------|---------|----------|-----------|-----------|----------|
| Qwen3.5-9B    | 62    | 62    | 58      | 0      | 0       | 0        | 263       | 263       | 263      |
| Gemma-3-12B   | 81    | 87    | 51      | 0      | 0       | 0        | 262       | 275       | 275      |
| Llama-3.1-8B  | 735   | 176   | 818     | 676    | 666     | 676      | 265       | 270       | 270      |
| Claude Opus 5 | 56    | 324   | 198     | 67     | 67      | 67       | 214       | 275       | 275      |

| Policy                  | Qwen3.5-9B |           | Gemma-3-12B |           | Llama-3.1-8B |            | Claude Opus 5 |            |
|-------------------------|------------|-----------|-------------|-----------|--------------|------------|---------------|------------|
|                         | trace      | both      | trace       | both      | trace        | both       | trace         | both       |
| Share all               | 0.00 [62]  | 0.00 [62] | 0.00 [87]   | 0.00 [87] | 0.00 [176]   | 0.00 [852] | 0.00 [324]    | 0.00 [391] |
| Majority vote           | 0.00 [62]  | 0.00 [62] | 0.26 [88]   | 0.26 [88] | 0.54 [149]   | 0.91 [757] | 0.69 [152]    | 0.73 [176] |
| LLM judge               | 0.00 [62]  | 0.00 [62] | 0.01 [87]   | 0.01 [87] | 0.59 [110]   | 0.77 [600] | –             | –          |
| Governance rule         | 0.84 [62]  | 0.84 [62] | 0.86 [65]   | 0.86 [65] | 0.86 [81]    | 0.93 [150] | 0.93 [103]    | 0.93 [103] |
| Independent-source vote | 0.69 [62]  | 0.69 [62] | 0.75 [65]   | 0.75 [65] | 0.80 [88]    | 0.92 [226] | 0.86 [110]    | 0.88 [120] |

## O The judges on both halves

On the 60 Live base configurations the Qwen judge promotes all 78 candidates and matches share-all’s damage of 147, and the Gemma judge promotes 253 of 258 and matches 142. The Llama judge and the frontier judge over Qwen halve damage, 72 against 145 and 78 against 147, and the frontier judge over its own agents holds 94 against 115 on the base configurations and under half the ceiling over all 100.

No judge matches the explicit rule’s high-stakes abstention. The local judge was not run over frontier agents, so judge and agent effects remain partly entangled.

Table 18 reports five combinations of judge and agent family on the 60 Live base configurations. Damage counts consumer reads while a false belief is retrievable, and the ceiling is share-all’s damage for that family. False and true writes use assertion grading. Writes reports promotions over candidates seen. Order sensitivity is the fraction of ten swapped pairs with different false or true write counts. Parse failures are responses naming no action.

Table 19 evaluates the selected prompt on four backbones. False-promotion rate is promotions among non-promote gold candidates, reported on the full test split and the majority-wrong subset. That subset holds the episodes on which a support-minus-contradiction proxy of the majority vote disagrees with the gold action, and on the frozen test split it is exactly the set of episodes whose gold action is not promote, so the two rates coincide by construction and the pre-registered comparison on it reduces to the full-set rate. FP/FN is false promotions divided by false withholdings, and values above one indicate over-promotion. Action shares report the four decisions, abstention recall uses high-stakes episodes, and accuracy is agreement with the gold action.

Table 18: Five LLM judges on CPB-Live over the 60 base configurations. Ceiling is share-all’s damage in the same agent family.

| Judge         | Agents        | Dmg↓ | Ceiling | FW↓ | TW↑ | Writes/cand. | Order sens. | Parse fail. |
|---------------|---------------|------|---------|-----|-----|--------------|-------------|-------------|
| Qwen3.5-9B    | Qwen3.5-9B    | 147  | 147     | 33  | 45  | 78/78        | 0.00        | 0           |
| Gemma-3-12B   | Gemma-3-12B   | 142  | 142     | 51  | 186 | 253/258      | 0.70        | 0           |
| Llama-3.1-8B  | Llama-3.1-8B  | 72   | 145     | 48  | 160 | 251/2002     | 0.20        | 0           |
| Claude Opus 5 | Qwen3.5-9B    | 78   | 147     | 17  | 34  | 51/88        | 0.20        | 0           |
| Claude Opus 5 | Claude Opus 5 | 94   | 115     | 24  | 54  | 229/359      | 0.80        | 2           |

Order sensitivity is the fraction of the ten candidate-swap pairs whose false and true write counts differed between the two orders. For the API judge it is confounded with sampling and descriptive.

Table 19: The same judge prompt on four backbones on the frozen Static test split of 400 episodes. FPR is the false-promotion rate, MW the majority-wrong subset, the episodes on which a support-minus-contradiction proxy of the majority vote disagrees with the gold action, which on the frozen test split is every episode whose gold action is not promote, so FPR on MW equals FPR, FP/FN false promotions over false withholdings, P/R/K/A the shares of the four actions, and the last two columns the abstain recall on the high-stakes family, where the governance rule scores 0.10, and the accuracy.

| Judge backbone | FPR↓  | FPR on MW↓         | FP/FN [CI]        | P/R/K/A              | ABSTAIN rec. | Acc.  |
|----------------|-------|--------------------|-------------------|----------------------|--------------|-------|
| Qwen3.5-9B     | 0.228 | 0.228 [0.17, 0.29] | 1.60 [1.02, 2.52] | 0.44/0.56/0.00/0.003 | 0.00         | 0.427 |
| Gemma-3-12B    | 0.285 | 0.285 [0.23, 0.35] | 2.92 [1.89, 4.89] | 0.50/0.40/0.10/0.000 | 0.00         | 0.448 |
| Llama-3.1-8B   | 0.228 | 0.228 [0.18, 0.28] | 1.33 [0.87, 2.03] | 0.42/0.27/0.31/0.000 | 0.00         | 0.477 |
| Claude Opus 5  | 0.077 | 0.077 [0.05, 0.11] | 0.23 [0.13, 0.36] | 0.22/0.24/0.54/0.000 | 0.00         | 0.412 |

## P Over-assertion of the judgment layer

Table 20 compares error direction on two tasks. The upper block evaluates same-proposition judgments against human labels on one hundred blind pairs and on the subset with annotator agreement. It reports Cohen’s kappa with cluster-bootstrap intervals and the ratio of false same-judgments to false different-judgments. The lower block reports false promotions divided by false withholdings on Static. Values above one indicate a bias toward same-proposition judgments or promotion, respectively. Error directions agree across models on annotation but differ on admission.

Table 20: Over-assertion of the judgment layer. Top: the same-proposition judgment against the human annotators, where the bias is false same-judgments over false different-judgments, so a value above one says same more often than the human. Bottom: the promotion direction against gold on CPB-Static, false promotions over false withholdings, so a value above one over-promotes.

| Model         | Reference               | $n$ | $\kappa$ [CI]     | Bias [CI]          | FP/FN |
|---------------|-------------------------|-----|-------------------|--------------------|-------|
| Qwen3.5-9B    | blind-100 (annotator A) | 100 | 0.36 [0.17, 0.54] | 2.86 [1.25, 10.00] | 20/7  |
| Qwen3.5-9B    | A/B-agreed subset       | 91  | 0.55 [0.32, 0.74] | 1.80 [0.57, 9.00]  | 9/5   |
| Claude Opus 5 | blind-100 (annotator A) | 100 | 0.40 [0.21, 0.56] | 5.25 [2.17, 24.00] | 21/4  |
| Claude Opus 5 | A/B-agreed subset       | 91  | 0.65 [0.44, 0.83] | 1.75 [0.50, 9.00]  | 7/4   |
| Qwen3.5-9B    | gold, CPB-Static test   | 400 | –                 | 1.60 [1.02, 2.52]  | –     |
| Gemma-3-12B   | gold, CPB-Static test   | 400 | –                 | 2.92 [1.89, 4.89]  | –     |
| Llama-3.1-8B  | gold, CPB-Static test   | 400 | –                 | 1.33 [0.87, 2.03]  | –     |
| Claude Opus 5 | gold, CPB-Static test   | 400 | –                 | 0.23 [0.13, 0.36]  | –     |

Human against human  $\kappa$  over all 100 pairs is 0.77; the agreed subset is the pairs on which the two annotators coincide. Intervals are cluster-bootstrap 95% intervals.

## Q The pre-registered comparisons

Copy-aware truth discovery and surface collapse decide identically on correlated-agreement episodes, the pre-registered null. On the majority-wrong subset the judge’s false-promotion rate exceeds the vote’s by

0.228, interval 0.169 to 0.288, Holm-corrected  $p < 0.001$ , and its false-promotion to false-withholding ratio is 1.60, interval 1.02 to 2.52. The governance rule beats the judge on high-stakes abstention recall by 0.10, interval 0.03 to 0.18, and review all correlated beats both collapse rules on accuracy by 0.11 while requesting evidence less often. A logistic regression and a gradient-boosted tree, fit on the Static training split over the view every rule receives, reach 0.645 and 0.653 accuracy against the best rule’s 0.368, a gap of about 0.28 with intervals excluding zero: the rules leave most of the structure unused. Both need a whole episode’s evidence graph, so they run on Static only. Table 21 gives the judge comparisons and their replications, where over-promotion holds on two of four judge backbones and reverses on the frontier model.

Table 21 reports the three judge-related comparisons among the six pre-registered tests. Qwen results are confirmatory, with Holm adjustment across all six, and other backbones provide descriptive replications. Comparison two measures false-promotion rate against majority vote on the majority-wrong subset. Comparison three assesses promotion bias through the FP/FN interval. Comparison four compares governance and judge abstention recall on high-stakes episodes. Appendix P gives the direction of the judges’ errors on annotation and on admission.

Table 21: The three pre-registered comparisons that involve the judge, the second, third and fourth of the six, Qwen3.5-9B confirmatory with Holm-adjusted  $p$  over the six, and the descriptive replication on three further judge backbones.

| # | Comparison                                               | Qwen (confirmatory)                       | Holm $p$ | Gemma             | Llama             | Opus 5            |
|---|----------------------------------------------------------|-------------------------------------------|----------|-------------------|-------------------|-------------------|
| 2 | Judge vs Majority vote, FPR on the majority-wrong subset | 0.228 vs 0.000; diff 0.228 [0.169, 0.288] | 0.000    | 0.285 vs 0        | 0.228 vs 0        | 0.077 vs 0        |
| 3 | Judge direction, FP/FN of PRO-MOTE (>1 over-promotes)    | 1.60 [1.02, 2.52]; sup-ported             | by CI    | 2.92 [1.89, 4.89] | 1.33 [0.87, 2.03] | 0.23 [0.13, 0.36] |
| 4 | Governance rule vs judge, AB-STAIN recall on R-STAKE     | 0.10 vs 0.00; diff 0.10 [0.03, 0.18]      | 0.012    | 0.10 vs 0.00      | 0.10 vs 0.00      | 0.10 vs 0.00      |

## R Reproduction materials

Code and data are at [https://github.com/lxy1134/iclr\\_2027](https://github.com/lxy1134/iclr_2027): the Static recipe with its adapters and gold rule, every Live scenario with its gold and lineage, the policies, the environment, the pipeline scripts that regenerate every table and figure from run summaries, the freeze files and the tool that verifies them. Transcripts and run summaries are released on acceptance. Table 22 lists the pre-registration freezes with their timestamps. One Live episode follows in full, then the four system prompts verbatim. The per-round user message of the agent lists its sources and the shared beliefs visible to it, the consumer’s lists the beliefs and the question, the judge’s the candidate, its cited sources and the visible beliefs, and the grader’s the false and true propositions and the sentence.

The public configuration, what the run sees, long lines folded for the page:

```
{
  "scenario_id": "live2-dc-00-2cca569c",
  "family": "D-CONC",
  "domain": "politics_governance",
  "n_agents": 6,
  "n_rounds": 6,
  "feeds": [
```

Table 22: The pre-registration freezes. Each artifact was hashed and its freeze file committed at the UTC time shown, before the runs it governs; the behaviour hash binds the artifact’s content and the source hash the files that produce it. The freeze files and the tool that verifies them ship with the code release.

| Artifact                       | Frozen, UTC         | Behaviour hash, first 16 hex |
|--------------------------------|---------------------|------------------------------|
| cpb-static-test-split          | 2026-09-09 14:27:02 | 3cc2607d164831ad             |
| gold-action-mapping-rule       | 2026-08-12 03:49:16 | 2d6e8d476bfb9f87             |
| pre-registered-comparisons-p14 | 2026-08-31 12:38:58 | 8ae0f8bc5504b040             |
| robustness-anchor-p12          | 2026-08-31 14:27:23 | 81cb4354f5c1283e             |
| semantic-gate-p4b              | 2026-08-17 13:53:22 | d5b5b9d01237bc89             |

```
{
  "source_id": "live2-dc-00-2cca569c-srcA",
  "text": "the Croften county board appointed Petra Kyle as its new
    head.",
  "source_type": "web_text",
  "round": 2,
  "agent": "a1"
},
{
  "source_id": "live2-dc-00-2cca569c-srcB",
  "text": "the Croften county board appointed Maro Dain as its new
    head.",
  "source_type": "web_text",
  "round": 2,
  "agent": "a2"
},
{
  "source_id": "live2-dc-00-2cca569c-auth",
  "text": "Local records confirm that the Croften county board
    appointed Petra Kyle as its new head.",
  "source_type": "register_document",
  "round": 4,
  "agent": "a4"
}
],
"task_queries": {
  "a5": "From shared knowledge, summarise the current facts on this
    topic (politics_governance)."
}
}
```

Its gold, read only at scoring:

```
{
  "scenario_id": "live2-dc-00-2cca569c",
  "false_texts": [
    "the Croften county board appointed Maro Dain as its new head."
  ],
  "true_texts": [
    "the Croften county board appointed Petra Kyle as its new head."
  ]
}
```

```

],
"swap_of": null,
"notes": "D-CONC: A is true (authoritative confirmation round 4); B
        false"
}

```

**Agent** src/agents/llm\_agent.py:

You are one agent in a small team sharing a memory of vetted beliefs. Each round you receive source documents assigned to you, and you can see the team's currently shared beliefs. Emit the candidate claims you would raise for admission to shared memory: factual, one sentence each. For every claim list the source ids it rests on, and, if it relies on a shared belief you read, that belief's id. Output STRICT JSON only: {"candidates": [{"text": "...", "sources": [...], "based\_on\_beliefs": [...]}]} Emit an empty list when nothing is worth raising.

**Consumer** src/cpb/consumer\_probe.py:

You are a member of a team that shares a memory of vetted beliefs. Answer the question using the shared beliefs below and nothing else. If they do not answer it, reply exactly: unknown. Reply with the answer only, in at most ten words.

**Judge** src/live\_env/live\_methods.py:

You are the admission judge for a team's shared memory. You are shown one candidate claim, the source documents it cites, and the beliefs already shared. Choose exactly one action: PROMOTE (admit it to shared memory), REQUEST\_EVIDENCE (defer pending better evidence), KEEP\_PRIVATE (do not admit it), or ABSTAIN. Reply with the single action word only.

**Assertion grader** scripts/24\_assert\_grader.py:

You judge whether one sentence, written by an agent in a team, asserts a given proposition as fact. Answer with one word.

## S Limitations

Every Live scenario is authored fiction, which is what lets the lineage be fixed and the truth be known, so the results say how the mechanism classes behave on this instrument and not how often the failures arise in a deployment. Parent independence in the correlated-agreement family is presumed from distinct citations, the dual-source sets show that collapse refuses copies and not that it admits genuine corroboration, and the protective effect of a competing truth is measured in the concurrent-conflict family alone. The store offers contest, demote and supersede each round and no compared policy uses them, so correction is untested rather than absent. The assertion grader and the judges are language models, both graders are validated against two human annotators on 120 items each, and the Live measurements are descriptive, with no hypothesis test. Beyond Mem0 and A-MemGuard, the comparison is over reimplemented mechanism classes.